Spatial Analysis of Train Station Directional Road Signage and Their Relationship to Train Stations Accessibility

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Supervisor: Dr. Liu Xin Tao Supervisor Signature: ____________________________

Abstract

Public transportation, such as trains, brings significant benefits to the community and the environment (Ezsias et al., 2023). Therefore, it is essential to guide people successfully and efficiently to access the train station. Train station directional signage can greatly assist in this regard (Jehle et al., 2022). This study aims to conduct a spatial analysis of directional signage leading to train stations and examine their relationship with pedestrian accessibility to reach the nearest entrance in Hong Kong, London, and Auckland. Various spatial analysis, including point data analysis and network analysis, are employed for this study. Key findings suggest that a balanced quantity, coverage area, and shortest travel distance of signage are essential, while number and placement of station entrances should serve as guidelines for determining these outcomes. Additionally, the study highlights the significance of distribution patterns, emphasizing the need for strategic placement. Moreover, centrality metrics such as degree, closeness, and betweenness offer valuable insights into signage effectiveness and route optimization. Overall, this study aims to provide valuable guidelines for designing an effective directional signage system, enhancing accessibility and navigation experience around train stations.

Introduction

Promoting the use of public train transportation has been an ongoing initiative to foster sustainability within the local environment and significantly benefit the community (Smith, 2003; Ezsias et al., 2023). In order to enhance the overall user experience and subsequently increase usage, various stakeholders, including the government collaborating with train companies, are committed to providing a comprehensive and optimized service accessible to everyone.

Efficient navigation around the station entrance is crucial as it ensures pedestrians have a seamless travel experience (Farr et al., 2012), thereby enhancing the overall usability and favourability of public transportation infrastructure in visited cities.

Directional signage at train stations plays a pivotal role in guiding pedestrians, passengers, and travellers by providing accurate directions to the station entrance from their original position (Jehle et al., 2022; Go & Taka, 2004). Moreover, these directional signages offer the shortest and most efficient routes, minimizing the time and distance required for passengers to access the station entrance (Jin et al., 2022). This not only prevents individuals from wandering aimlessly on the streets but also facilitates quick entry into the station, even in adverse weather conditions such as intense sunlight or rainy days. Ultimately, train station directional signage contributes significantly to fostering a positive perception of the transportation system.

Understanding the significance of train station directional signage for passenger navigation underscores the equal importance of optimizing their placement. This optimization involves determining the appropriate quantity, specifying the coverage area, and managing the density of these directional signages. Additionally, considering the spatial distribution, patterns, and other spatial characteristics of these signages around the station entrance is essential for enhancing wayfinding and overall station navigation. This comprehensive approach to signage planning aids in improving accessibility.

Furthermore, this study seeks to conduct an in-depth spatial analysis by comparing stations within the same cities as well as between different well-known cities globally. The objective is to uncover the reasons behind varying levels of accessibility in directional signages, exploring why some cities excel while others lag behind. This comparative approach aims to identify specific spatial factors that may influence the results, shedding light on why travellers may take longer to reach a station entrance in certain locations and shorter times in others.

The ultimate goal is to determine an optimized solution for enhancing the placement of entire train station directional signage systems. By understanding the spatial dynamics and factors contributing to efficient wayfinding, the study aims to propose strategies that can be universally applied to improve accessibility, streamline navigation, and enhance the overall commuter experience. Through this comparative analysis, valuable insights can be gained to inform the development of standardized and effective directional signage systems for train stations worldwide.

Accessibility

Accessibility encompasses various aspects to ensure a seamless transportation experience (Jehle et al., 2022). One key facet is pedestrian access to the station from the original position, emphasizing the ease with which individuals can reach a railway station from their starting point (Jehle et al., 2022), be it their homes or another location. The second aspect involves pedestrian access within the station's platform (Jehle et al., 2022; Zheng et al., 2007; Go & Taka, 2004), concentrating on how individuals navigate within the station premises. Another critical dimension of accessibility is station-to-station connectivity (Zheng et al., 2007). Having addressed and understood various accessibility aspects, it is important to clarify that this study specifically targets the accessibility of pedestrians from their starting point to the station entrance. The spatial analysis will specifically concentrate on studying and evaluating the effectiveness of signages placed on the streets, primarily designed to guide pedestrians rather than vehicles.

Directional Signages

The primary purpose of signs is to convey a concise message intentionally within a limited space. As a result, signs adhere to numerous standards and serve various purposes. While signs may not necessarily include directional information, directional signs typically provide messages guiding individuals on how to reach specific destinations. Even within the same category of train station directional signs, some studies may identify banners, maps, papers affixed to light poles or walls, as well as boards installed on streets, all serving as signages (Zheng et al., 2007).

It is noteworthy and essential to distinguish between the terms "signs" and "signages" in this study. Signs can be created by anyone, such as shops using signs to guide pedestrians or cafes displaying signs atop their establishments. On the other hand, signages are generally installations on streets and roads by local authorities (Transport Department, 2022; Health and Safety Executive, 2009), functioning as government or train station company assets to guide the public without targeting specific user types. Regarding directional signages, they are constructed not solely for pedestrians but also for vehicles and vessels at various locations.

In the context of this study, train station directional signage specifically focuses on those displaying the train station company logo along with directional messages. There are various types of directional signage that we aim to document. One category includes signages without specific directional information; nevertheless, they are considered noteworthy due to their towering height, standing approximately three to four meters taller than typical directional signs. These landmarks serve as distinctive features, strategically positioned close to the station entrance. Although lacking explicit directional messages, their prominent placement aids pedestrians and commuters in easily identifying the station entrance. Notably, individuals from a distance can use these towering signages as reference points to locate the station and navigate towards it. The tall signages without explicit directional guidance prove effective due to their proximity to the station entrance, providing a clear visual cue for those approaching the landmark. In contrast, the general directional signs stand around two meters tall and feature the train station company logo alongside directional instructions incorporated into the signage. Another prevalent type, commonly observed in the city centre of London and also present in Hong Kong and Auckland, involves the installation of information boards with maps. These boards prominently display the train station name accompanied by the company logo, such as Transport for London. This approach, distinct from traditional signage, offers comprehensive information in a centralized format.

These signages are intentionally positioned to facilitate navigation, align with the natural direction of passenger flow, and minimize confusion. The goal is to guide passengers efficiently to the platform. For the purpose of this study, train station directional signs will be referred to in different forms, primarily as "signs," but also as "signages" and "directional signs," all conveying the same meaning.

Study Area

In Hong Kong, public transport is indispensable (Loo et al., 2010). The city, with a mere territory of approximately 1114km^2 and a densely populated 7.5 million residents (Lands Department, 2023; Census and Statistics Department, 2023), relies heavily on efficient transportation systems. Unlike in larger countries such as New Zealand, owning a car in Hong Kong is not as practical due to the city's small size and relatively low-income levels (Cullinane, 2002). To effectively address the transportation needs of a large population in such a confined space, public transportation options like the MTR play a crucial role. This not only proves more efficient than private vehicles in the context of Hong Kong but also addresses the economic challenges associated with car ownership (Cullinane, 2002). Moreover, the MTR provides a cost-effective means of travel for citizens (Tang & Lo, 2010), contributing to fewer individuals opting for private vehicles. The affordability and convenience of the MTR system not only promote widespread usage but also mitigate issues related to traffic congestion, as the railway system remains unimpeded. In conclusion, in Hong Kong, the MTR is a common and integral mode of transportation, serving as a practical and efficient solution in the face of geographical constraints and economic considerations.

Tsim Sha Tsui and Shau Kei Wan stations differ significantly. Tsim Sha Tsui caters to travellers and visitors, renowned for its shopping, dining, cultural attractions like museums, and hosting Hong Kong's tallest building (HK Tourism Board, 2024). It is a vibrant, cosmopolitan area with high-end malls, hotels, and entertainment venues, targeting popular tourist spots such as Victoria Harbour, Harbour City Mall, and the Avenue of Stars (HK Tourism Board, 2024). Tsim

Sha Tsui serves as a well-connected hub, offering various transportation options, including ferries to Hong Kong Island. Hong Kong Tsim Sha Tsui station is distinctive as it effectively combines two stations, Tsim Sha Tsui and Tsim Tung, sharing the same platform and passage (MTR, 2023). Passengers can seamlessly access or interchange between Tsim Sha Tsui and Tsim Tung without incurring extra charges, enhancing the convenience of travel for those using both stations. In contrast, Shau Kei Wan is primarily a residential area surrounded by local markets and traditional shops. It has a more local and less touristy ambiance compared to Tsim Sha Tsui. Shau Kei Wan is connected to the tram, MTR network, and buses, facilitating convenient access to different parts of Hong Kong Island. Given the distinct urban layouts, varying levels of foot traffic, and differing degrees of tourist attractions between Tsim Sha Tsui and Shau Kei Wan, it is anticipated that the spatial placement of train station signages would be well-adapted to these specific environments.

Auckland is a large city with a population density lower than that of Hong Kong and London (Davis, 2018). Notably, the majority of Auckland residents heavily rely on cars rather than public transport (Jiang, et al., 2024). The New Zealand government, along with its citizens, prioritizes environmental conservation, particularly in Auckland (Auckland Council, 2018), the largest city in the country. Consequently, Auckland has seen relatively slow progress in the extensive and large-scale construction of railways and underground subways (Auckland Transport, 2014). The railway network in Auckland is relatively simple, with fewer networks and less complexity. An interesting aspect is the significant reliance on cars for commuting within the city, as it takes a considerable amount of time to travel by foot from one location to another. The accessibility of the closest station platform is not always convenient and might involve a lengthy walk. Aucklanders tend to rent cars for city and country road trips rather than using public transportation like railways (Hickman et al., 2014). This trend extends to both tourists and locals, as private cars dominate the traffic landscape. Moreover, Auckland's train development has lagged behind when compared to cities like Hong Kong and London. This developmental contrast is reflected in the inadequate and less comprehensive directional signages at train stations. The government appears to place less emphasis on providing efficient public transportation services, specifically in the realm of train services. This disparity results in a less-than-optimal public transport system in Auckland.

Britomart, situated in Auckland's bustling central business district, stands as a prominent transport hub encompassing not only a pivotal train station but also functioning as a key bus interchange and ferry terminal (Atkinson et al., 2022). The station is seamlessly connected to various train lines, including the Southern, Eastern, and Onehunga lines. Renowned for its upscale fashion boutiques, eclectic dining options, and meticulously restored historic architecture, Britomart offers a vibrant urban experience (Atkinson et al., 2022). The locale frequently hosts diverse events such as art exhibitions, pop-up markets, and live performances, adding to its dynamic atmosphere. Nestled close to the waterfront, Britomart invites leisurely strolls along the harbour, allowing one to appreciate the scenic beauty of Auckland (Atkinson et al., 2022). Given its status as the central business district, Britomart primarily caters to commuters, workers, tourists, visitors, university students and residents who rely on its efficient train services. In contrast, Papatoetoe serves predominantly as a suburban station within a residential area, fostering a more community-oriented and authentic experience for local residents. The train services primarily cater to residents, commuters, and students from the surrounding suburban areas.

London, the capital and largest city of England, with a population of approximately 8.8 million (Roskams, 2022) and is renowned for its well-developed public transportation services. Given the city's high population density, comparable to that of Hong Kong (Davis, 2018), residents and travellers are inclined to rely on public transport rather than private vehicles (Balcombe, et al., 2005; GLA, 2014). The escalating costs of petrol, particularly amid recent wars and world conflicts, further encourage a shift towards public transportation. Although the ownership of private vehicles is not as infrequent as in Hong Kong, individuals residing outside the central areas of London are less dependent on public transport. The intricacy and complexity of London's train station network surpass that of Hong Kong, yet it efficiently connects people to most destinations without extensive walking. Interestingly, London holds the distinction of being the first city to construct a train and initiate passenger services over 180 years ago (Network Rail, 2020). This early adoption of railway transport has instilled a lasting concept of eco-friendly travel through public transport, particularly trains. The cost-effectiveness of train tickets in London incentivizes a greater number of people to opt for train services over owning a private vehicle. This choice not only proves economical but also aids in alleviating traffic congestion and mitigating the challenges of finding parking spaces in the bustling city environment. Overall, London's well-established and eco-conscious public transport infrastructure significantly influences the commuting habits of its population.

Charing Cross Station, situated in Central London near Trafalgar Square and the West End, functions as a transportation hub and a sought-after tourist destination (Bremner, et al., 2024). Positioned amidst iconic landmarks like the National Gallery, Covent Garden, and other popular attractions (Bremner, et al., 2024), it serves as a gateway to London's vibrant theatres and shopping districts. The station facilitates both national and international train services, providing access to various parts of London and the United Kingdom. On the other hand, Albany Park Station, nestled in a residential suburb, offers a more localized and tranquil experience. While it may not attract travellers, the station caters to the commuting needs of residents in its suburban surroundings. Albany Park Station facilitates the daily journeys of local residents, students connecting them to work, school and other destinations within the suburban setting.

Figure 1: Directional signage at Charing Cross and Albany Park Stations, London

Figure 2: Directional signage at Tsim Sha Tsui and Shau Kei Wan Stations, Hong Kong

Figure 3: Directional signage at Britomart and Papatoetoe Stations, Auckland

Literature Review

Signage

Zheng et al. (2007) discovered that, in Japan's interchange stations, there tends to be more directional signs at further distances from the stations compared to closer distances. The results also indicated that pedestrians traveling from a greater distance are more likely to utilize maps or signs. However, the opposite trend was observed for travellers from shorter distances to the station, where travellers need to make decisions along the routes, especially at intersections. This suggests that the information sign system needs enhancement by providing a greater number of signs. Interestingly, different studies adopt varied approaches to defining signs. In Zheng et al.'s (2007) research, the definition is based on their locations, encompassing outside entrances, aisles toward the stations, and inside stations. The study also describes various forms of station directional signs, which include posters attached to wall poles, directional signs, orientation maps, signs attached along stairs, upper hanging signs, installed standing directional sign poles, and even signs hanging on electricity poles.

The literature conducted by Brons et al. (2009) validates the significance of access – referring to how passengers' access and navigate train stations, including factors such as station design, signage, and facilities for individuals with mobility challenges – in influencing rail use. This influence is established through an analysis of factors that affect passengers' satisfaction with rail journeys. Furthermore, Brons et al. (2009) provides evidence that enhancing the journey to the station is crucial. This is because under budget constraints, investing in access improvements might be more feasible than enhancing the level of rail service. Such investments prove to be a more cost-effective approach and have the potential to attract new passengers. In addition, Brons et al. (2009) recommends a policy shift towards investing in access to the rail service, taking into account the likely costs involved. This suggests a need for a strategic change in policy focus, emphasizing the importance of improving accessibility, aligning with the findings that well-designed access facilities and assets such as signage contribute significantly to passenger satisfaction and increased rail usage.

Jehle et al. (2022) present a comprehensive exploration in two dimensions of accessibility: pedestrian access to the station from the original position and pedestrian access within the stations. Jehle et al. (2022) aims to understand perceived pedestrian accessibility to railway stations from the original position. Key considerations include the importance of comfort, traffic safety, and security as determinants of perceived pedestrian accessibility. Factors such as freedom from barriers, availability of street crossings, and adequate lighting emerge. Within the Stations, perceived weaknesses in accessibility arise due to minor issues, such as broken street lights or less-than-ideal underpasses, exerting an influence on individuals' choices to walk to the station. In Jehle et al.'s (2022) study, a social approach is adopted, providing insights into demographic and geographic factors influencing the decision to walk to railway stations. Pedestrian age and the size of the city play pivotal roles, suggesting that different age groups and urban contexts demand tailored approaches to enhance accessibility. The study reveals diversity in needs and abilities among different user groups based on age, luggage, daytime, and weather conditions (Jehle et al., 2022). Lastly, Jehle et al. (2022) underscore the necessity of incorporating all four accessibility components—transportation, land use, temporal, and individual—for a comprehensive evaluation of perceived accessibility.

While this project aims to evaluate the spatial pattern of directional road signage impacting accessibility to train stations, Sun et al. (2022) take a different approach from studying environmental psychology and behaviour to analyse the relationship between visual behaviour, signs, and the environment, aiming to improve wayfinding efficiency. Leveraging eye-tracking technology for transportation building wayfinding, they visualize data on decision points to identify navigation errors. In parallel, the studies by Zheng (2020), Jing & Bai (2020), and Tu et al. (2019) focus on accessibility within stations, emphasizing the importance of clearer signage, effective landmarks, and improved wayfinding programs for optimal navigation.

Espera et al. (2015) conducted a study analyzing the signage along major roads in the Filipinos, emphasizing the crucial role these signs play in guiding and instructing individuals in their daily road activities. Espera et al. (2015) highlight the significance of effective sign placement for pedestrians and emphasize that careful consideration must be given to the positioning of warning or directional signs to ensure the safety of pedestrians who may be unfamiliar with the road. Proper sign placement should allow pedestrians sufficient time for perception, identification, decision-making, and execution of necessary actions. Espera et al. (2015) stress the importance of adhering to guidelines and standards, such as those outlined in the Manual on Uniform Traffic Control Devices, to enhance visibility and convey the intended meaning to pedestrians. Furthermore, Espera et al. (2015) highlight the significance of maintaining uniformity in colour, size, and placement across signage to ensure standardization. The issue of inattentiveness among Filipinos, who often use a lack of attention as a common excuse after missing important details in various circumstances, is also addressed.

Methodology

In order to analyse spatial patterns and deviations from random expectation in various species distributions across different ecological contexts, Clark and Evans (1954) use the nearest neighbour approach in their study. They apply this method to examine the spatial distribution patterns of grassland forbs (Lespedeza capitata Michx., Liatris aspera Michx., and Solidago rigida L.) as well as the distribution of trees in an oak-hickory woodlot. The distances to the nearest neighbour evaluate the level of aggregation or uniformity in the spatial patterns of these populations successfully. Statistical methods are then used to assess the significance of departure from random expectation.

Vincent et al. (2022) employ Ripley's K-function to precisely measure the size and structure of spatial molecular and atomic clustering in materials. Vincent et al. (2022) extends the application of Ripley's K-function to 3D point patterns, particularly emphasizing its effectiveness in Atom Probe Tomography data analysis. Through machine learning models based on metrics derived from Ripley's K-function, the method demonstrates improved accuracy in estimating cluster properties, highlighting the significance of Ripley's K-function for reliable characterization of clustering in material systems.

Haase (2009) explores the application of spatial pattern analysis, specifically focusing on Ripley's K-function in plant ecology. The study describes the process of field data acquisition, statistical analysis, and testing for the null hypothesis of complete spatial randomness (Haase, 2009). Different methods of edge correction for Ripley's K-function are evaluated through computer-generated random patterns and real distribution patterns of Mediterranean shrubland. Haase (2009) recommends the unbiased estimator proposed by Ripley, to enhance the reliability and comparability of results in spatial pattern analysis.

Leong and Sung's (2015) exploration of spatial patterns in spatio-temporal crime analysis contributes to the understanding of crime dynamics. This literature review focuses specifically on their discussion of spatial patterns. Leong and Sung (2015) recognize that spatial events associated with crime are not uniformly distributed across maps; instead, they exhibit clustering and absence in different areas. They introduce the concepts of hotspots and collocation, emphasizing the significance of spatial clustering techniques for analysing point events. Hotspots, defined as geographic areas with higher-than-average incidences, are analyzed using clustering method, including kernel density approach, to transform spatial point patterns into smooth images, creating hotspot maps.

Mclafferty et al. (2000) used a spatial statistical method, Kernel Density Estimation (KDE), for crime analysis to identify crime hotspots in Brooklyn, New York. This was done by using maps to pinpoint areas with a high concentration of crimes, generating a smooth surface that represents the variation in crime density over space. The concept of hotspots extends to both specific crime types and areas with an elevated overall risk (Mclafferty et al., 2000). The research by Mclafferty et al. (2000) aimed to target crime control efforts in these hotspot areas, leading to a reduction in reported crimes.

Le et al. (2022) conducted a study to identify traffic accident hotspots in Hanoi, Vietnam. They employed an integrated approach, combining GIS-based techniques with Kernel Density Estimation (KDE) and spatial autocorrelation analysis to pinpoint hazardous locations on roads based on the severity of accidents. Statistical significance was assessed using Moran's I statistic indices, and the hotspots were organized according to their significance levels. The study by Le et al. (2022) successfully demonstrated that this approach effectively identified traffic accident hotspots with statistical significance. Furthermore, Le et al. (2022) validated that GIS and KDE are powerful tools for analysing point data specific to traffic accidents, providing a visual representation that aids investigations into the underlying causes of these incidents.

As this project aims to assess the importance of directional signages using centrality measures, specifically degree, closeness, and betweenness centrality, it seeks to integrate Vignery and Laurier's (2020) insights into our methodology for analysis. Vignery and Laurier (2020) proposed an integrated methodology for evaluating centrality indices within a real friendship student network. Their study emphasizes centrality measures in understanding the importance of nodes within networks. The results reveal six latent dimensions of centrality, including the ability to reach friends, transfer and receive information, significance for the network's structure, connections with prestigious friends, degree of cross-connectivity, and position as a confluent node (Vignery & Laurier, 2020).

Summary

The literature review for the project focusing on the spatial pattern and analysis of train station places nearby stations reveals a gap in research regarding the relationship between directional signs and train station accessibility. The forms of signs under consideration in this study are those placed on the road network, outside, and near station facilities, however existing studies vary in their approaches to analysing directional signs, with most focusing on signs within the station (Zheng et al., 2007; Jing & Bai, 2020; Zheng, 2020, Tu et al., 2019), while few explore signs from the original location to the station (Jehle et al, 2022). Unlike other studies that delve into comprehensive spatial analyses covering time, distance, physical, social, and psychological aspects (Zheng et al., 2007; Jehle et al., 2022; Sun et al., 2022), there is a scarcity of research adopting a holistic spatial analysis approach. Some researchers, emphasize the design of signs (Dewar & Pronin, 2023; Department for Transport, 2015; Department for Transport et al, 2019), while others focus on the orientation and physical placement on roads (Farr et al., 2012; Zhang et al., 2010; Pietrantonio, 2014).

In the literature on signs, researchers commonly concentrate on road signs rather than pedestrian signs. While Espera et al. (2015) specifically discuss road signs, certain aspects can also be applicable to pedestrian signs. Furthermore, concerning the placement of signs, existing literature predominantly addresses road signs (Babic et al., 2022; Khalilikhah & Heaslip, 2016), with a notable lack of analyses on the placement of pedestrian signs.

Moving to the methodology section, the review notes a deficiency in spatial analysis methods for train station directional signs. Kernel Density Estimation (KDE), commonly used in traffic accidents or crime analysis (Yamada & Thill, 2004; Leong & Sung, 2015; Mclafferty et al., 2000), lacks relevant studies in the context of train station directional signs. On the other hand, Average Nearest Neighbour and Ripley's K function, applied in epidemiology, ecology, biology, physics, and sociology, have limited cases related to train station directional signs (Clark & Evans, 1954; Haase, 2009; Melyantono et al., 2021; Vidanapathirana et al., 2022). Centrality measures, various fields including those used in social networks, logistics, aviation, and maritime shipping networks, have been explored (Kuang et al., 2022; Vignery & Laurier, 2020). The literature suggests a need for tailored spatial analysis methods for directional signs, as existing methods may not directly apply due to differences in study objects.

Research Methodology

Data Collection

To collect spatial data on signage in the city of Hong Kong, spatial data for traffic aid signage was accessed from the Hong Kong CSDI government website (CSDI & Transport Department, 2021). This dataset includes all pedestrian and vehicle signs. After downloading the shapefile and loading it into the ArcGIS Pro software, layers were removed to retain only pedestrian signs, specifically directional signages. However, the spatial data does not explicitly indicate whether it includes directional signage for train stations, identifiable by the presence of a Hong Kong MTR logo. Consequently, any observed MTR logo directional signs were marked by the process of traversing every street in Tsim Sha Tsui Station and Shau Kei Wan Station. Streets were only visited within 500 meters from the station entrance. The zone was differentiated through buffer analysis in ArcGIS Pro. The dataset for Hong Kong's road network and station entrances was retrieved from the Hong Kong CSDI (CSDI & Transport Department, 2021) and ArcGIS Open Data (Esri China (HK) Ltd, 2022), respectively.

To collect spatial data on signs in the city of Auckland, the process mirrors that used in Hong Kong. Data was obtained from the Auckland Transportation website (Auckland Transport, 2022), encompassing both traffic and pedestrian signs. Unlike the method employed in Hong Kong, where signs were distinguished through walking and personal observation, Google Street View in Google Maps (Google, n.d.) was utilized to determine if a sign indicated an Auckland train station logo. If affirmative, the sign was marked. Similarly, streets were only visited within 500 meters from the station entrance. Auckland's road network and station entrance dataset were acquired from Auckland Transport (Auckland Transport, 2018; Auckland Transport, 2022).

When collecting spatial data on signs in the city of London, the dataset for train station directional signs was not available. Instead, another approach was employed to determine if a signage contained train station directional signage featuring the TFL (Transport for London) logo. Google Street View in Google Maps (Google, n.d.) was utilized to examine every street around Charing Cross Station and Albany Park Station within 500 meters. Whenever a TFL train station signage was observed, a new point is created in ArcGIS pro, associated with the corresponding position shown on Google Maps. The dataset for the road network and station entrance in London was sourced from the Ordnance Survey Data Hub (Ordnance Survey Data Hub, 2023) and ArcGIS Open Data (EsriUK_EducationTeam, 2021), respectively.

Data Analysis

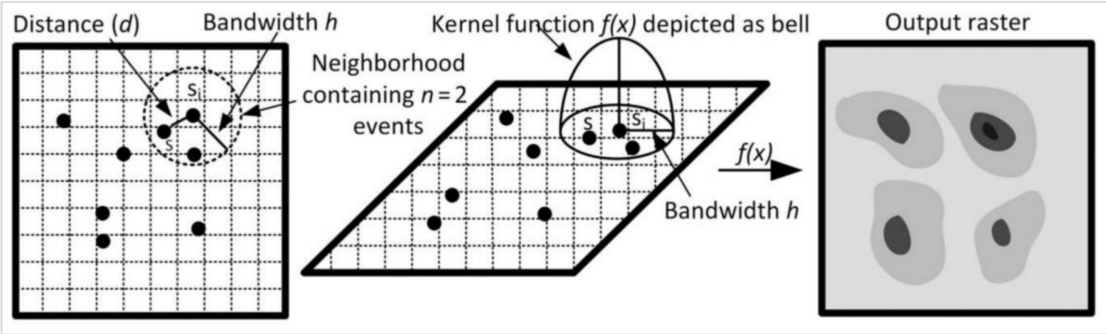
Esri ArcGIS Pro is the primary tool for this project's analysis. In the **spatial analysis** component, the distance of the shortest path and its associated time, quantity of signage, its ratio with entrance and road, and coverage area will serve as indices for spatial analysis using Esri ArcGIS Pro. The first and most straightforward method involves counting the number of signages at each study station. Additionally, the associated ratios between signage and each station entrance, as well as the ratio between signage and road, will be calculated, representing proportions and density. The second method measures distance in meters and time in minutes. It determines the duration and distance it takes for people to move from directional signs to station entrances virtually. This is done by calculating the shortest path from each signage to the nearest station entrance, generated using ArcGIS Pro. For coverage area, a convex hull will be adopted to calculate the coverage area (Kirkpatrickf & Seidel, 1986).

To comprehend the spatial distribution and patterns of directional road signages around train station entrances, three **point data analysis** approaches will be employed: Kernel Density Estimation (KDE), Average Nearest Neighbourhood (ANN), and Ripley's K Function. Each signage will be represented as a point in the dataset with corresponding coordinates.

Kernel Density Estimation (KDE), as a nonparametric technique, means it estimates the density of point features rather than their values. It serves as a method to analyse and visualize the intensity of point distributions, representing the random variable of directional road signages. This allows for the identification of any clustering around station entrances on a continuous surface representation on a map (Grekousis, 2020). The method calculates the density of directional road signages in each square on a grid (Cai et al., 2012), particularly searching a 100-meter radius around the station entrances. Conceptually, KDE portrays directional road signages as smooth hills on a map with the highest points indicating their locations and the hills gradually flatten as one moves away. This identifies clusters of directional road signages on the map, distinguishing areas with varying concentrations (He et al., 2018). Additionally, the nature breaks classification method uses 10 classes, employing the Jenks Natural Breaks Optimization method to discern natural groupings in the data distribution for a more granular analysis. As ArcGIS applies the quadratic kernel function as described by Silverman (Silverman, 1986), the functional form is as shown in equation.

$$\hat{f}(x) = \frac{1}{nh^2\pi}\sum_{i=1}^{n}\left(1 - \frac{d^2}{h^2}\right)^2$$

Where:
h = bandwidth
n = number of directional roads signages inside the search radius
d = the distance between the central event s_i and the s event inside the bandwidth h.



(Grekousis, 2020)

The Average Nearest Neighbours (ANN) analysis, a statistical test, will be employed to quantify the degree of clustering or dispersion of directional signages. It helps determine whether the signages exhibit a regular, random, or clustered pattern for wayfinding. ANN calculates the average distance to the nearest neighbour for each signage in the overall spatial arrangement (Grekousis, 2020). In ANN analysis, the results are represented in a ratio: a value less than 1 indicates a tendency toward clustering, while a value larger than 1 indicates a dispersed pattern. A value close to 1 reflects a random pattern. The test outputs include the observed mean distance, the expected mean distance, the nearest neighbour ratio, the p-value, and the z-score. The p-value represents the probability that the observed point pattern results from complete spatial randomness. A smaller p-value suggests it is less likely that the observed pattern is generated by complete spatial randomness. If the p-value exceeds the significance level, the null hypothesis cannot be rejected (Grekousis, 2020). It is calculated based on the formula below (Clark & Evans, 1954 & Grekousis, 2020).

$$Ratio = \frac{observed\ mean\ distance}{expected\ mean\ distance} = \frac{\bar{d}_{min}}{E(d)} = 2\bar{d}_{min}\sqrt{n/a}$$

$$\bar{d}_{min} = \frac{\sum_{i=1}^{n} d_{min}\,(s_i)}{n}$$

$$E(d) = 1/\left(2\sqrt{n/a}\right)$$

Where:

d_{min} = the average nearest neighbours’ distance for directional signages' spatial pattern
$d_{min}(si)$ = the distance of directional signages s_i to its nearest neighbor
n is the total number of directional signages
$E(d)$ = The expected mean nearest neighbour distance under complete spatial randomness.
a is the coverage area of the directional signages

The approach of Ripley’s K function analysis is different from the nearest neighbour approach. While the nearest neighbour looks at distances to the closest events, the K-function checks spatial patterns across various distances. It considers all distances between directional signages and finds the best models to describe spatial patterns in a study area (Bailey & Gatrell, 1995; Fotheringham & Zhan, 1996). The calculation involves considering small distance steps of 100 meters, each centred around a signage, and then determining the average number of directional signages within each of these small distances. When the observed value of the K function is larger than the expected value K for a range of distances, the distribution is more clustered than a random distribution. Conversely, when the observed value of the K function is smaller than the expected value K for a range of distances, the distribution is more dispersed than a random distribution. When the observed value of the K function is closer to the expected value K, the distribution is random (Grekousis, 2020). It is calculated based on the formula below (Grekousis, 2020). The upper and lower confidence envelopes aim to signify complete randomness in the distribution of directional signages. They are employed to assess the significance of the results. The directional signages will undergo random distribution 999 times at a confidence level using the Monte Carlo approach (Grekousis, 2020).

$$K(r) = \frac{a}{n^2} \sum_{i=1}^{n} \sum_{j=1, i \neq j}^{n} \frac{I_r(r_{ij})}{w_{ij}}$$

Where:
$K(r)$ = the expected number of directional signages inside the radius (r),
a = area of region encompassing all (n) directional signages
n = number of directional signages
I_r = indictor function where value of I_r equal to 1 when the distance between directional road signages i and j is shorter than h; otherwise, it equals 0.
w_{ij} = weights used for edge correction and is calculated as the fraction of the circle's outer boundary, with a radius (r) centered at a point, that lies within the study area.

In this study, **network analysis** will be employed, using signages as point data represented as nodes within a larger network of road networks. The primary objective is to assess the significance of signages, their specific locations, and explore the underlying reasons for their importance. Three centrality measures will be utilized to provide a comprehensive analysis: degree centrality, betweenness centrality, and closeness centrality (Vignery & Laurier, 2020).

Degree centrality counts how many connections neighbours of a node have (Vignery & Laurier, 2020). It is used to identify nodes with many direct road contacts. For this study, it will quantify the number of roads connected to each signage node. The first step involves snapping the nodes to the road network and determining if the signage is situated at an intersection indicating pedestrian activity. Utilizing Esri ArcGIS Pro, the number of roads connected to each signage will be computed. Higher degree centrality values indicate a greater level of connectivity or importance within the local road network (Vignery & Laurier, 2020). Furthermore, signages with higher degree centrality values are predicted to be more likely positioned at key intersections, hubs, or areas with a high volume of road connections, potentially making them more influential in guiding or directing pedestrians.

Closeness centrality measures how close a node is to other nodes in the entire network, evaluating nodes based on the lowest sum of the shortest paths and time required to reach all other nodes (Vignery & Laurier, 2020). Since this study will compare signages in different-sized road networks, the closeness centrality score needs to be normalized. This is achieved by first inversing of the sum of the shortest path distance values then multiplying the previous score with the maximum possible edges, which is (n-1), where n is the number of nodes (Vignery & Laurier, 2020). A node with a normalized closeness of 1 is directly connected to all nodes in the network. For this study, the shortest paths of each signage to every station entrance will be measured, and the signage with the lowest sum of the shortest path distances will be identified. This will be done by employing network analysis functions in ArcGIS Pro to determine the shortest path from each signage to every station entrance. The objective is to identify the top three signages with the shortest average distance to all station entrances at the station, highlighting their importance and influence in closeness centrality. Utilizing the shortest time and distance to reach all station entrances, we calculate the sum to determine higher closeness centrality values and take the inverse of result. These values indicate that these signages can reach all other station entrances in the road network most quickly, facilitating effective broadcast of directional indicators (Vignery & Laurier, 2020). This is particularly helpful in the diffusion process when a large number of people are seeking station entrances in these areas.

Betweenness centrality captures the role of a node as a bridge between other groups of nodes by measuring how many shortest paths pass through a node (Vignery & Laurier, 2020). For this study, the first step involves pairing the signages with their closest station entrances. Secondly, Esri ArcGIS Pro will be used to generate the shortest paths from each signage to its nearest station entrance. Thirdly, the study involves calculating the number of shortest paths passing through each signage. Lastly, since the study compares signages in road networks of different sizes, the betweenness centrality score needs to be normalized. This normalization takes into account the potential for larger networks to yield higher maximum scores. The normalization formula is shown below (Vignery & Laurier, 2020). A higher betweenness centrality suggests that an individual signage has a greater influence on potentially controlling and intermediating pedestrian flow within a road network. Signages with relatively higher betweenness centrality have a greater impact on disrupting the guidance or direction of pedestrians if they are removed too (Vignery & Laurier, 2020).

$$\frac{(n-1) \times (n-2)}{2}$$

Where:
n = number of nodes

Results and Analysis

Figure 4: Hong Kong – Tsim Sha Tsui

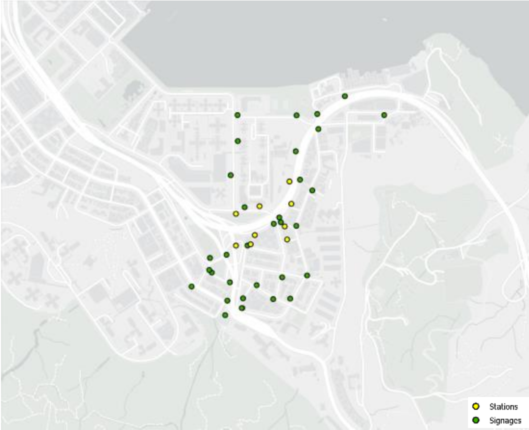


Figure 5: Hong Kong – Shau Kei Wan

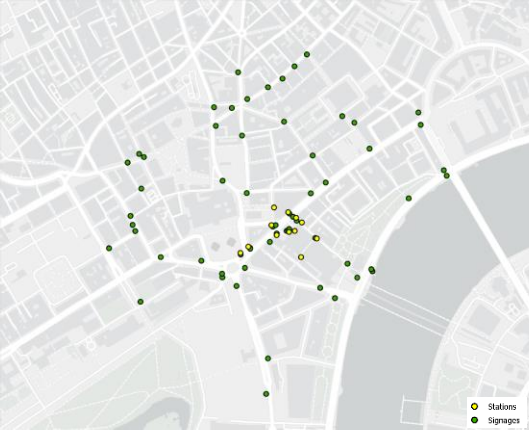


Figure 6: London – Charing Cross

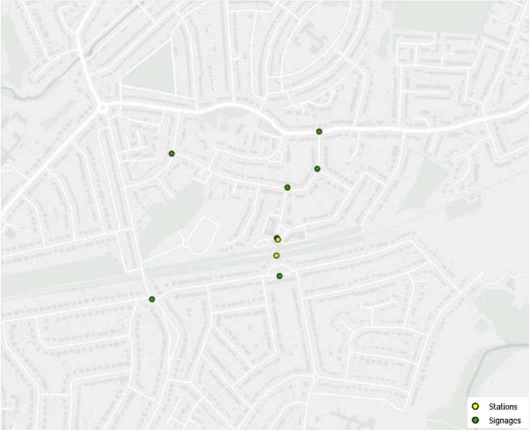


Figure 7: London – Albany Park

Figure 8: Auckland – Britomart

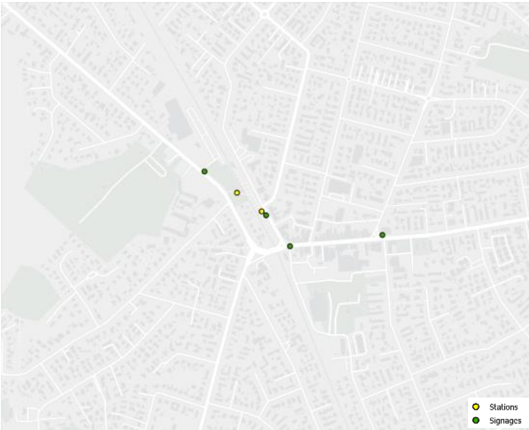


Figure 9: Auckland – Papatoetoe

The six maps depict the distribution of directional signages at train stations in Hong Kong, London, and Auckland. The maps share a common scale of 1:6000 and utilize a light grey base background from Esri ArcGIS Pro, displaying roads, parks, buildings, houses, and water bodies. Urban stations for tourists and locals are shown on the left, while less central or residential stations are displayed on the right.

Upon visual inspection, it's evident that Hong Kong has more directional signages than the other cities, both in popular urban stations and less central residential stations. The signs are dispersed in Hong Kong, with a higher concentration near station entrances. In contrast, London and Auckland show a slight concentration of signs near entrances. Additionally, signs in all stations are notably positioned at road intersections or junctions rather than the middle of the road. Examining station entrances, Hong Kong again stands out with a higher number in both popular urban and less central residential stations. The entrances are also more distributed in Hong Kong compared to other cities. While these observations provide an initial insight, accurate distribution analysis will require further statistical and spatial analysis in subsequent sessions.

Quantity and Ratio

City	Station	Station Entrance	Signs	Ratio
Hong Kong	Tsim Sha Tsui	22	81	1: 3.68
	Shau Kei Wan	9	33	1: 3.67
London	Charing Cross	12	61	1: 5.08
	Albany Park	2	7	1: 3.5
Auckland	Britomart	7	33	1: 4.71
	Papatoetoe	2	4	1: 2

Table 1: Number of Train Station Directional Signs and Stations Entrances in Each Study Station

City	Station	Road	Signs	Ratio
Hong Kong	Tsim Sha Tsui	364	81	1: 0.22
	Shau Kei Wan	243	33	1: 0.14
London	Charing Cross	322	61	1: 0.19
	Albany Park	155	7	1: 0.05
Auckland	Britomart	391	33	1: 0.08
	Papatoetoe	119	4	1: 0.03

Table 2: Number of Train Station Directional Signs and Road in Each Study Station

Now focusing solely on the number of station entrances, signages, and roads, let's examine the data for different stations. In Hong Kong, Tsim Sha Tsui station has the highest number of train station directional signages with 81, and it also has 22 station entrances. This results in a ratio of 1:3.68 for signages to entrances, meaning that each station entrance has an average of 3.68 signages guiding pedestrians. Shau Kei Wan station in Hong Kong has 33 signages and 9 entrances, giving it a ratio of 1:3.67, which is similar to Tsim Sha Tsui station. Moving on to London, Charing Cross station has the second-highest number of train station directional signages with 61, along with 12 station entrances. Albany Park station, on the other hand, has only 7 signages and 2 entrances, resulting in a ratio of 1:5.08 and 1:3.5, respectively. This reveals that each Charing Cross station entrance has an average of 5.08 directional signages, which is the highest proportional value compared to the other studied stations. In Auckland, Britomart station has 33 train station directional signages and 7 station entrances, while Papatoetoe station has 4 signages and 2 station entrances. This gives a ratio of 1:4.71 and 1:2, respectively. Papatoetoe station has the least proportional relationship between station entrances and directional signages, as each entrance only has 2 signages guiding pedestrians towards it.

After examining the relationship between station entrances and signages, let's focus on the proportion between roads and signages. It's important to note that the post-processed spatial data of roads were only counted if they were within 500 meters of the station entrances zone. This was achieved by using a pre-process clipping function in ArcGIS Pro software to remove roads that were located beyond the 500-meter range. In Hong Kong, Tsim Sha Tsui station has 364 roads within the designated area, while Shau Kei Wan has 243 roads. This results in a ratio of 1:0.22 and 1:0.14, indicating that each road has an average of 0.22 and 0.14 signages placed along it, respectively. Tsim Sha Tsui station has a higher proportion and density of signages compared to the other studied stations. In London, Charing Cross station has 322 roads within the designated area, while Albany Park has 155 roads. This results in a ratio of 1:0.19 and 1:0.05, respectively. For Auckland, Britomart station has 391 roads within the designated area, while Papatoetoe has 119 roads. This gives a ratio of 1:0.08 and 1:0.03, respectively. Similar to the ratio between station entrances and signages, Papatoetoe station has the least proportional relationship between roads and directional signages. Each road in Papatoetoe only has 0.03 signages guiding pedestrians towards the station entrances.

Coverage Area

City	Station	Coverage Area of Signs (m^2)
Hong Kong	Tsim Sha Tsui	709337.29
	Shau Kei Wan	230876.42
London	Charing Cross	638195.60
	Albany Park	181649.89
Auckland	Britomart	374283.09
	Papatoetoe	36052.64

Table 3: Area Coverage of Train Station Directional Signs Using Convex Hull Method

Finally, the investigation focused on the coverage area of signages within each station. The coverage area was determined using the convex hull approach calculated in ArcGIS Pro software. Despite Hong Kong being the smallest city compared to others, it boasts the largest coverage area by directional signages. The combined coverage of Tsim Sha Tsui and Shau Kei Wan stations is 940,213.71 m^2, with Tsim Sha Tsui individually covering 709,337.29 m^2, and Shau Kei Wan covering 230,876.42 m^2. In London, directional signages at Charing Cross Station cover 638,195.60 m^2, while Albany Park covers 181,649.89 m^2. Conversely, in Auckland, directional signages at Britomart cover 374,283.09 m^2, and Papatoetoe has the smallest coverage area, covering 36,052.64 m^2.

Shortest Distance and Time

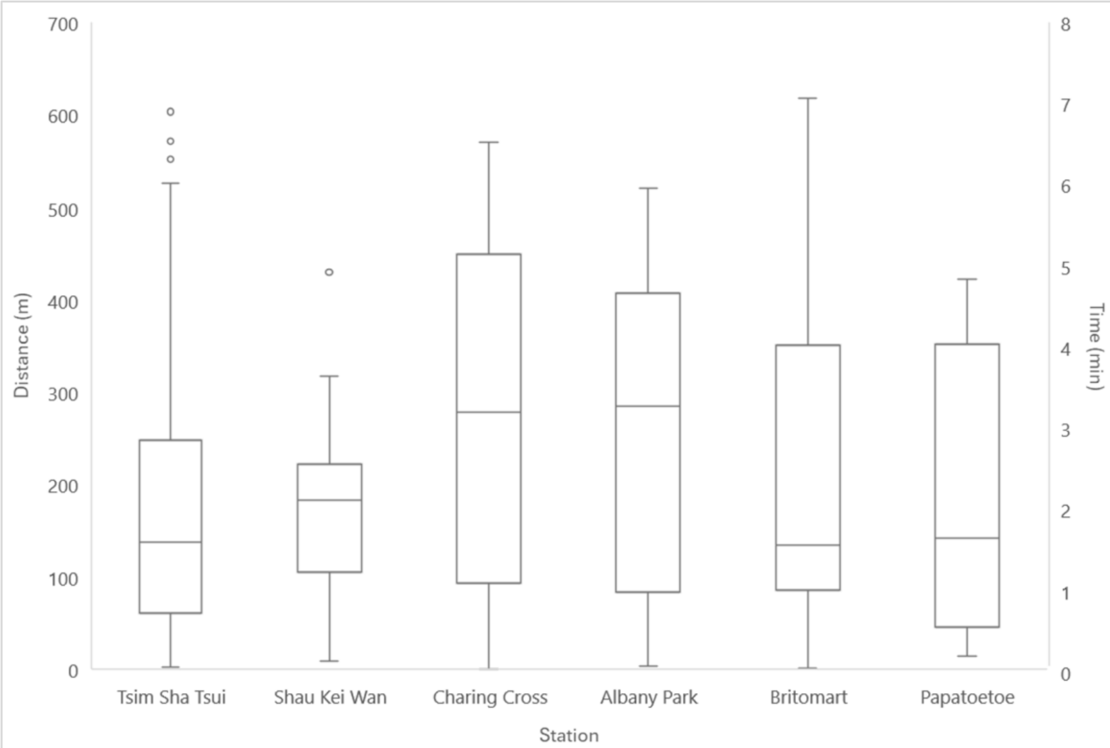


Figure 10: Shortest Distance and Time from Each Sign to Its Closest Station Entrance for Each Study Stations.

	Tsim Sha Tsui	Shau Kei Wan	Charing Cross	Albany Park	Britomart	Papatoetoe
Maximum	526.24	317.61	570.35	520.53	618.55	422.16
3rd Quartile	248	222.03	449.56	406.87	350.84	352.41
Median	137.65	182.76	277.97	284.48	133.98	142.02
1st Quartile	61.27	104.88	92.90	84.00	85.52	45.73
Minimum	2.21	8.82	0.20	3.84	1.55	14.02

Table 4: Box and Whisker Plot Statistics for Shortest Distance from Directional Signages to Closest Station Entrances.

	Tsim Sha Tsui	Shau Kei Wan	Charing Cross	Albany Park	Britomart	Papatoetoe
Maximum	6.31	3.81	6.84	6.25	7.42	5.07
3rd Quartile	2.98	2.66	5.89	4.88	4.21	4.23
Median	1.65	2.19	3.34	3.41	1.61	1.70
1st Quartile	0.74	1.26	1.11	1.01	1.03	0.55
Minimum	0.03	0.11	0.00	0.05	0.02	0.17

Table 5: Box and Whisker Plot Statistics for Time from Directional Signages to Closest Station Entrances

The distribution of the shortest distance from directional signages to the closest station entrances varies across different locations, as indicated by the box-and-whisker plot statistics. In Tsim Sha Tsui, the distances are relatively concentrated around the median of 137.65, with a smaller Interquartile Range (IQR) from 61.27 to 248. The maximum distance of 526.24 indicates the presence of outliers on the higher end, reaching up to 603.57, suggesting a positively skewed distribution with a tail extending towards longer distances. Shau Kei Wan exhibits a narrower IQR (104.88 to 222.03) around the median distance of 182.76. The maximum distance of 317.61 and the distribution of values suggest a negatively skewed pattern, indicating that there is one location with significantly longer distances (429.78) from directional signages to station entrances. Charing Cross displays a normal distribution, with a median distance of 277.97 and an IQR spanning from 92.90 to 449.56. The maximum distance recorded is 570.35. In Albany Park, the distribution is slightly negatively skewed with a median distance of 284.48 and an IQR from 84 to 406.87. The maximum distance of 520.53 suggests some locations with notably longer distances, contributing to the skewness observed in the distribution. Britomart's distribution, with an IQR (85.52 to 350.84) around the median distance of 133.98, includes a maximum distance of 618.55. This contributes to a positively skewed distribution, indicating a tail on the upside with longer distances. Papatoetoe shows a distribution with a wider IQR (45.73 to 352.41) around the median distance of 142.02. The positively skewed pattern is evident, with the maximum distance reaching 422.16, indicating shorter distances from directional signages to station entrances. In summary, the distributions across these studied stations exhibit both positive and negative skewness, implying that some distances are concentrated on the lower end, while other stations have notably longer distances, leading to the elongation of the upper tails in the box-and-whisker plots. As time is associated with the shortest path distance, the scale of the statistical results from the box-and-whisker plot will remain consistent.

Kernal Density Estimation

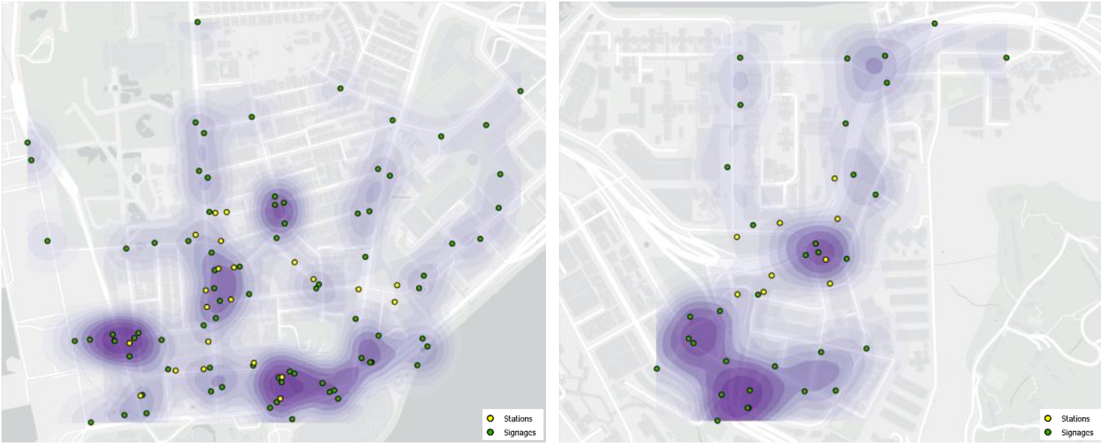


Figure 11: Kernal Density Estimation of Tsim Sha Tsui (Left) and Shau Kei Wan (Right) Train Station Directional Signages in Hong Kong

Figure 12: Kernal Density Estimation of Charing Cross (Left) and Albany Park (Right) Train Station Directional Signages in London

Figure 13: Kernal Density Estimation of Britomart (Left) and Papatoetoe (Right) Train Station Directional Signages in Auckland

The kernel density estimation map provides insights into the spatial distribution and concentration of directional signages around the studied station. It highlights areas of both higher and lower density, allowing for the identification of signage clusters.

On the left side of the Figure 11, representing Tsim Sha Tsui train station, the KDE map reveals a noticeable concentration of directional signages in proximity to shopping malls, hotels, outdoor leisure zones, entertainment, and artistic venues, as well as a diverse array of shops. The KDE analysis further demonstrates a distinct concentration of signages along Nathan Road and Salisbury Road leading towards the station's main entrance. Notably, there is a higher density of signage along major streets frequented by people for various purposes such as shopping, sightseeing, dining, or attending events, contrasting with the fewer signages along less significant roads lined with commercial buildings. Furthermore, a significant density of directional signs is noted in the vicinity of key transportation hubs, including Mody Road and Middle Road bus hubs and terminus, and taxi stands. Remarkable variations in density become apparent as one moves away from the station entrance, indicating shifts in the prominence of directional signages in different areas. Shifting focus to the right side of Figure 11, which represents Shau Kei Wan train station's outdoor areas, there are slight differences compared to Tsim Sha Tsui. In Shau Kei Wan, the KDE analysis illustrates an evenly distributed low density of directional signs along residential streets. Similar to Tsim Sha Tsui, however, the analysis reveals an increased density of signages near a central entrance, markets, various shops, grocery stores, and central bus stations.

The KDE analysis revealed distinctive patterns at London's Charing Cross station, where a particular area shows clustering near the station entrances, while the remaining signages displayed an even density level. Notably, a significant cluster of signages is observed along Cranbourn Street and Long Acre, adjacent to St Martin's Courtyard and TK Maxx, a prominent shopping mall and department store, respectively. Along the same street, Leicester Square station and Covent Garden station are also present. In contrast, Albany Park station exhibits varying density scales due to the limited number of signages. However, it is noteworthy that a dense concentration of signage surrounds the station entrance, as well as at the intersection of the major road where buses travel.

In Britomart, directional signage exhibits a clustered pattern primarily around commercial buildings, the ferry terminal, the event venue – The Cloud, the waterfront area, and the surrounding hotels and restaurants. Additionally, clusters are evident near station entrances. Some scattered directional signages are also found along Queen Street, a major shopping street. Notably, there is a significant area lacking directional signage, particularly in the right-middle section of the map (Figure 13), which encompasses a mix of commercial, residential, and leisure areas. In Papatoetoe, directional signage appears in a sparser and more dispersed arrangement, with only four signages visible. The limited number of points accentuates a higher density level near the station entrance. The other two signages are located near a roundabout and an intersection of road networks within a residential area.

Average Nearest Neighbour

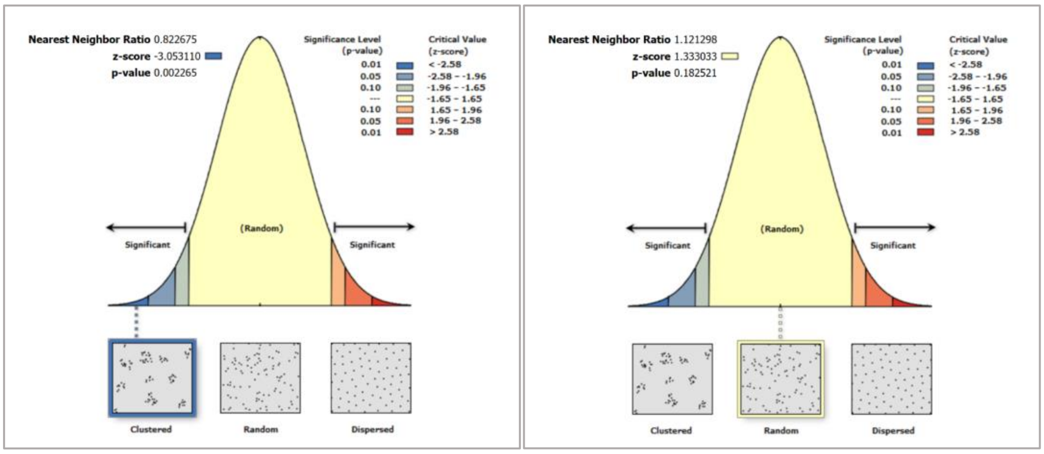


Figure 14: Average Nearest Neighbour Analysis for Train Directional Signs at Tsim Sha Tsui Station (Left) and Shau Kei Wan (Right) in Hong Kong

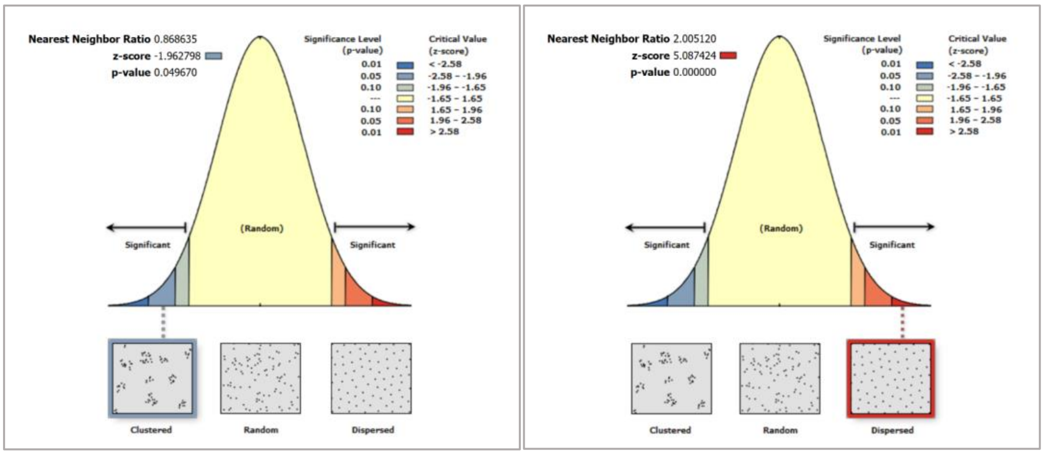


Figure 15: Average Nearest Neighbour Analysis for Train Directional Signs at Charing Cross Station (Left) and Albany Park (Right) in London

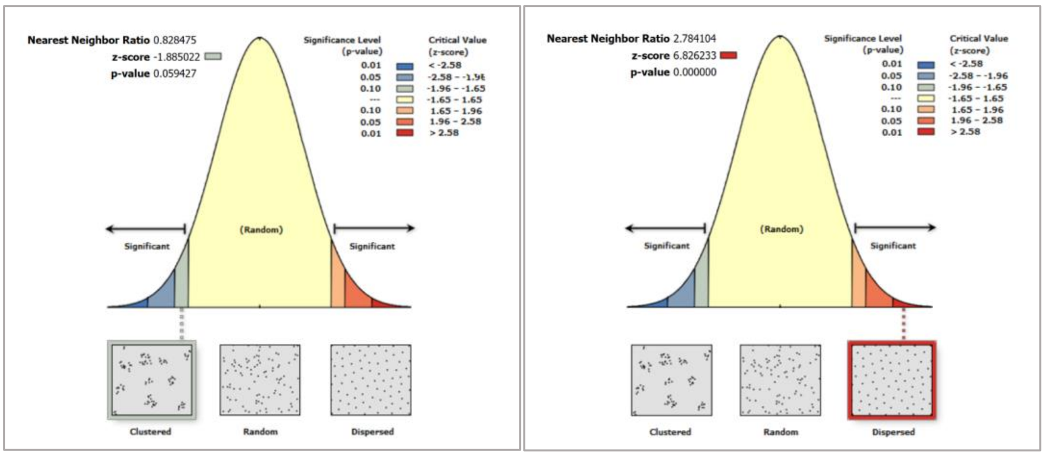


Figure 16: Average Nearest Neighbour Analysis for Train Directional Signs at Britomart Station (Left) and Papatoetoe (Right) in Auckland

	Tsim Sha Tsui	Shau Kei Wan	Charing Cross	Albany Park	Britomart	Papatoetoe
Observed Mean Distance	44.5788	53.5788	48.5144	187.6470	50.5112	186.8999
Expected Mean Distance	54.1876	47.7828	55.8514	93.5839	60.9689	67.1311
Nearest Neighbour Ratio	0.8227	1.1213	0.8686	2.0051	0.8285	2.7841
z-score	-3.0531	1.3330	-1.9628	5.0874	-1.8850	6.8262
p-value	0.0023	0.1825	0.0497	0.0000	0.0594	0.0000

Table 6: Table of Average Nearest Neighbour Analysis for Train Directional Signs at Various Study Stations

At Tsim Sha Tsui station, the observed mean distance between signs is 44.5788 meters, which is lower than the expected mean distance of 54.1876 meters, indicating a clustered distribution. This is supported by a Nearest Neighbour Ratio (NNR) of 0.8227, with a corresponding z-score of -3.0531 and a significant p-value of 0.0023, suggesting a non-random distribution. Contrarily, at Shau Kei Wan station, the observed mean distance (53.5788 meters) is higher than the expected mean distance (47.7828 meters), yielding an NNR of 1.1213. While the z-score (1.3330) indicates a deviation from randomness, the p-value (0.1825) does not reach statistical significance, implying a more random distribution compared to Tsim Sha Tsui.

Charing Cross station exhibits a similar pattern to Tsim Sha Tsui, with a clustered distribution of signs (NNR = 0.8686) and a significant z-score (-1.9628) accompanied by a p-value of 0.0497. Albany Park and Papatoetoe stations show extreme dispersion, indicated by high NNR values of 2.0051 and 2.7841 respectively. Both stations have significantly high z-scores (5.0874 and 6.8262) with p-values close to zero, demonstrating highly non-random distributions. Lastly, Britomart station also displays significant clustering (NNR = 0.8285) with a z-score of -1.8850 and a p-value of 0.0594, indicating a tendency towards non-random distribution, although not as pronounced as in Albany Park and Papatoetoe.

Ripley's K-function

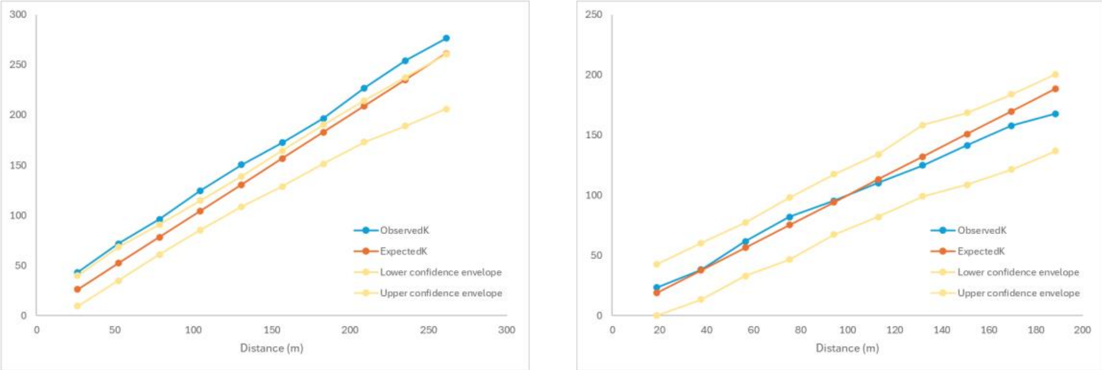


Figure 17: Multi-Distance Spatial Cluster Analysis (Ripley's K-function) for Spatial Distribution of Train Directional Signs at Tsim Sha Tsui Station (Left) and Shau Kei Wan (Right) in Hong Kong

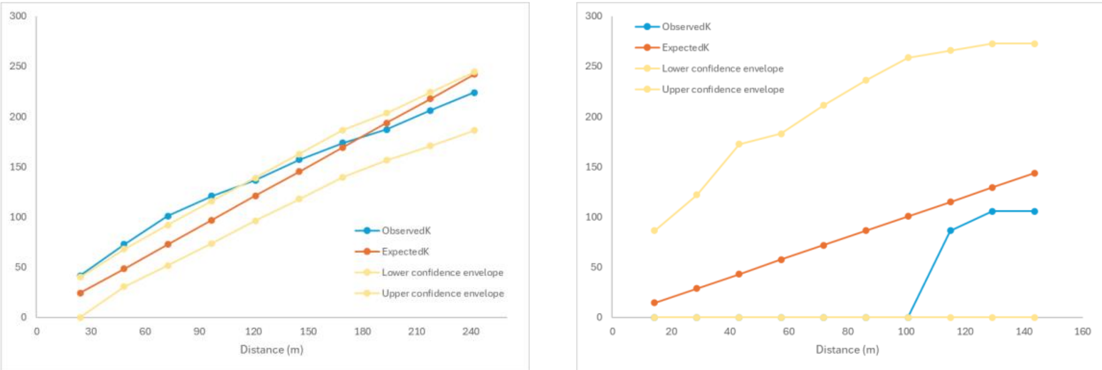


Figure 18: Multi-Distance Spatial Cluster Analysis (Ripley's K-function) for Spatial Distribution of Train Directional Signs at Charing Cross Station (Left) and Albany Park (Right) in London

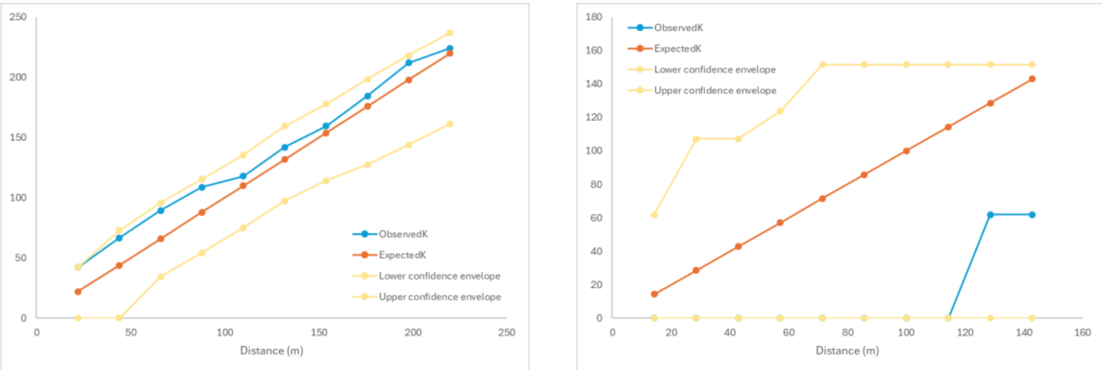


Figure 19: Multi-Distance Spatial Cluster Analysis (Ripley's K-function) for Spatial Distribution of Train Directional Signs at Britomart Station (Left) and Papatoetoe (Right) in Auckland

Hong Kong - Tsim Sha Tsui				
Distance (m)	ObservedK	ExpectedK	Lower confidence envelope	Upper confidence envelope
26.1329	43.2355	26.1329	9.66774	39.8611
52.2658	71.6979	52.2658	34.8575	68.3613
78.3987	96.1928	78.3987	61.1442	90.6915
104.532	124.56	104.532	85.3832	114.798
130.665	150.395	130.665	108.52	138.758
156.797	172.401	156.797	128.621	164.352
182.93	196.472	182.93	151.324	190.187
209.063	226.729	209.063	172.942	214.223
235.196	254.135	235.196	188.954	237.402
261.329	276.504	261.329	205.993	260.491
Hong Kong - Shau Kei Wan				
18.8544	23.3468	18.8544	0	42.6253
37.7087	38.1252	37.7087	13.4793	60.2812
56.563	61.7699	56.563	33.0174	77.4326
75.4174	81.9913	75.4174	46.6936	98.1307
94.2717	95.313	94.2717	67.3964	117.51
113.126	110.333	113.126	81.9913	134.117
131.98	125.002	131.98	99.0521	158.346
150.835	141.372	150.835	108.674	168.356
169.689	157.771	169.689	121.314	183.833
188.543	167.816	188.543	136.8	200.384

London - Charing Cross				
Distance (m)	ObservedK	ExpectedK	Lower confidence envelope	Upper confidence envelope
24.1937	41.4858	24.1937	0	39.8583
48.3875	72.771	48.3875	30.4423	68.071
72.5812	100.966	72.5812	51.4569	92.0488
96.7749	120.677	96.7749	73.675	115.635
120.969	136.627	120.969	96.267	139.029
145.162	156.922	145.162	117.902	162.721
169.356	173.738	169.356	139.504	186.598
193.55	186.952	193.55	156.5	203.564
217.744	206.149	217.744	170.663	224.295
241.937	224	241.937	186.243	244.352
London - Albany Park				
14.3715	0	14.3715	0	86.2205
28.7431	0	28.7431	0	121.934
43.1146	0	43.1146	0	172.441
57.4861	0	57.4861	0	182.901
71.8576	0	71.8576	0	211.196
86.2291	0	86.2291	0	236.125
100.601	0	100.601	0	258.662
114.972	86.2205	114.972	0	265.75
129.344	105.598	129.344	0	272.653
143.715	105.598	143.715	0	272.653

Auckland - Britomart				
Distance (m)	ObservedK	ExpectedK	Lower confidence envelope	Upper confidence envelope
21.9818	42.1288	21.9818	0	42.1288
43.9637	66.6115	43.9637	0	72.9692
65.9455	89.3687	65.9455	34.398	95.7601
87.9274	108.776	87.9274	54.3881	115.374
109.909	117.911	109.909	74.9688	135.425
131.891	141.827	131.891	97.2923	159.497
153.873	159.497	153.873	114.085	177.908
175.855	184.439	175.855	127.551	198.349
197.837	212.044	197.837	143.897	218.231
219.818	224.248	219.818	161.341	237.072
Auckland - Papatoetoe				
14.2933	0	14.2933	0	61.849
28.5867	0	28.5867	0	107.126
42.88	0	42.88	0	107.126
57.1733	0	57.1733	0	123.698
71.4667	0	71.4667	0	151.499
85.76	0	85.76	0	151.499
100.053	0	100.053	0	151.499
114.347	0	114.347	0	151.499
128.64	61.849	128.64	0	151.499
142.933	61.849	142.933	0	151.499

Table 7: Table of Ripley's K-function Analysis for Spatial Distribution of Train Directional Signs at Various Study Stations

At Tsim Sha Tsui Station, the observed K-values consistently exceed the expected K-values across various distance intervals, indicating a clustering of train directional signs. This clustering is evident as the observed K-values are notably higher than the expected K-values, suggesting that the signs are closer together than would be expected by chance alone. Additionally, the upper confidence envelope consistently lies above the expected K-values, reinforcing the presence of significant clustering. The signs appear to cluster particularly strongly within shorter distances, gradually tapering off as distance increases. Conversely, at Shau Kei Wan Station, the observed K-values generally exceed the expected K-values, suggesting some degree of spatial aggregation. However, they are not consistently higher across all distance intervals, as seen in Tsim Sha Tsui Station. Furthermore, the upper confidence envelope consistently surpasses the expected K-values, which might initially imply statistically significant clustering. However, this pattern may suggest a more random distribution rather than a clear clustering trend. The signs at Shau Kei Wan seem to display a mix of stronger clustering at shorter distances and a lack of clear clustering as distance increases, hinting at a potentially random spatial arrangement rather than a consistent clustering pattern.

The KDE analysis indicates at Charing Cross Station, the observed K-values surpass both the expected K-values and upper confidence envelope, pointing to a statistically significant clustering of train directional signages at shorter distances. This clustering is evident in the higher observed K-values compared to the expected, indicating a closer proximity of signages than would be expected. However, it's important to note that as distances increase, the observed K-values start to fall below the expected K-values, suggesting the presence of significant dispersion as well. This suggests a pattern where while signages do cluster notably at shorter distances, there's also a gradual trend towards dispersion as distance increases. Conversely, the KDE analysis of train directional signages at Albany Park Station presents a starkly different picture. Here, the observed K-values remain consistently low across all distance intervals, often at zero. This indicates a lack of significant clustering and suggests a more dispersed or random spatial distribution of the signages. Unlike Charing Cross Station, where signages clustered closely together, at Albany Park Station, the absence of significant clustering indicates a more scattered arrangement of directional signages. This is further supported by the upper confidence envelope, which remains substantially higher than the observed K-values, indicating a lack of statistically significant clustering.

At Britomart Station, the analysis reveals a clear clustering pattern of train directional signages. The observed K-values consistently exceed the expected K-values but fall below upper confidence envelope across various distance intervals. This suggests that while there is a clustering tendency, it may not be statistically significant beyond certain distances. The signages at Britomart Station appear to be closely grouped together, with a tendency for stronger clustering within shorter distances gradually diminishing as distance increases. In contrast, Papatoetoe Station presents a markedly different spatial distribution pattern compared to Britomart Station, but it exhibits a similar spatial distribution pattern compared to Albany Park Station. Here, the observed K-values remain consistently low across all distance intervals, often at zero. This indicates a lack of clustering and suggests a more dispersed or random spatial distribution of the signages. The absence of statistically significant clustering is evident as the observed K-values consistently fall below both the expected K-values. While Britomart Station boasts closely clustered signage, Papatoetoe Station reveals a noticeably dispersed layout of directional signages.

Degree Centrality

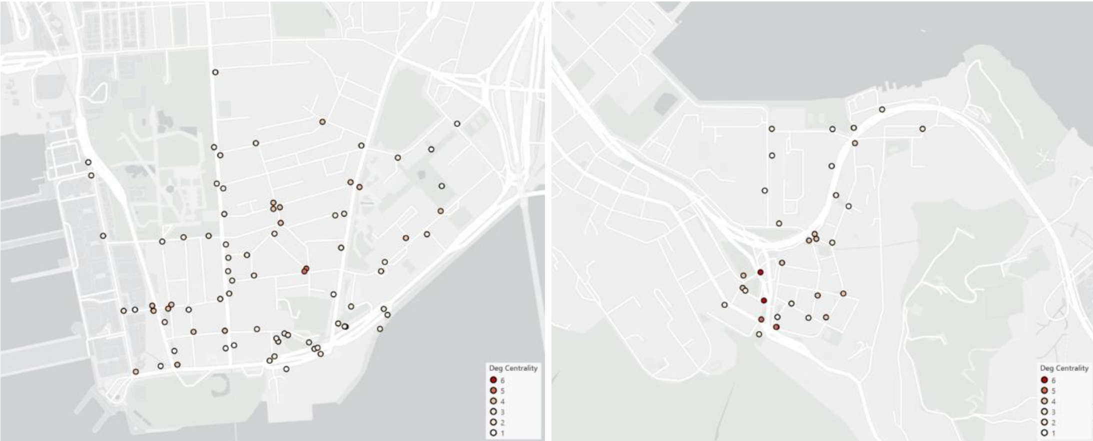


Figure 20: Degree Centrality Analysis of Train Directional Signs at Tsim Sha Tsui Station (Left) and Shau Kei Wan (Right) in Hong Kong

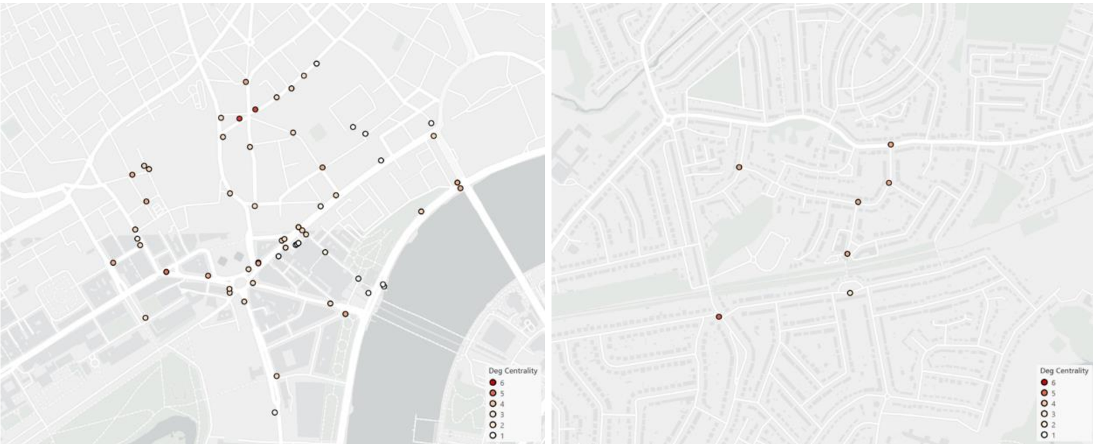


Figure 21: Degree Centrality Analysis of Train Directional Signs at Charing Cross Station (Left) and Albany Park (Right) in London

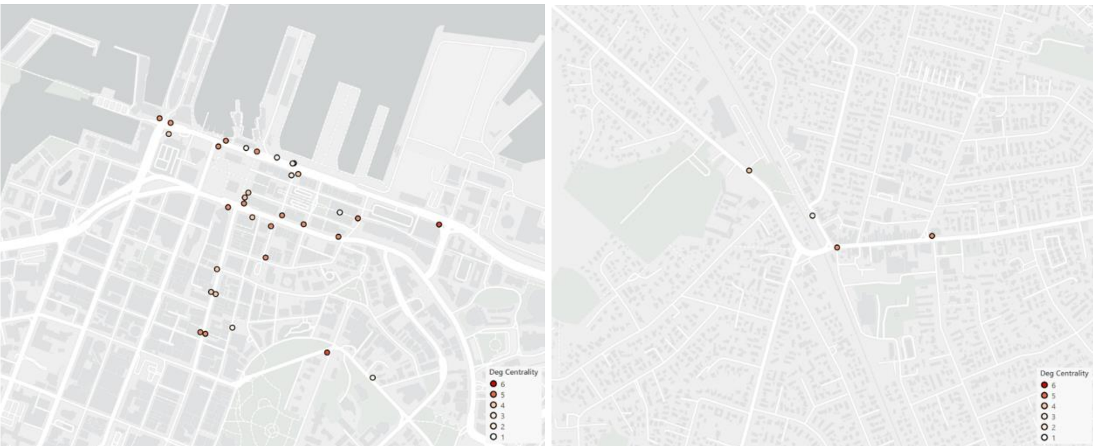
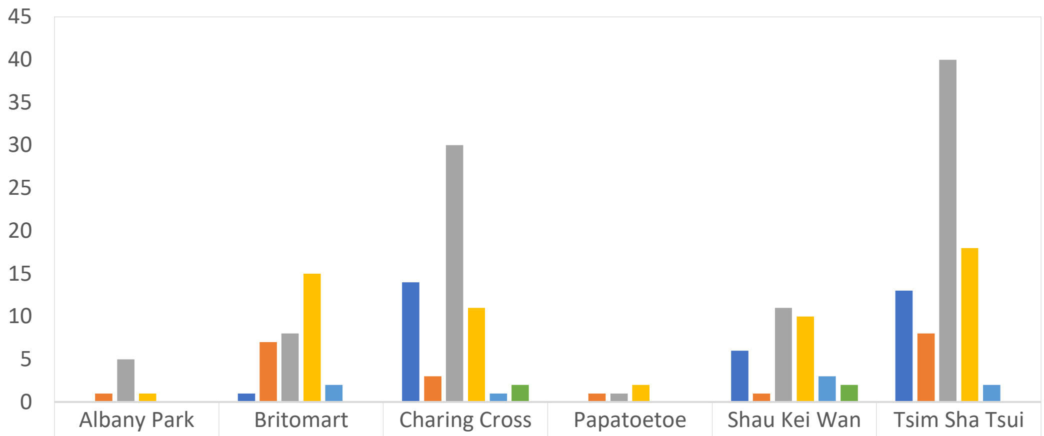


Figure 22: Degree Centrality Analysis of Train Directional Signs at Britomart Station (Left) and Papatoetoe (Right) in Auckland



	Albany Park	Britomart	Charing Cross	Papatoetoe	Shau Kei Wan	Tsim Sha Tsui
1		1	14		6	13
2	1	7	3	1	1	8
3	5	8	30	1	11	40
4	1	15	11	2	10	18
5		2	1		3	2
6			2		2	

Figure 23: Clustered Column Graph and Table of Degree Centrality Analysis Result of Train Directional Signs for Each Study Station

The results of the degree centrality analysis are depicted on maps from Figure 20 to 23, where varying degrees of centrality are represented by colour intensity. Higher degrees of centrality are depicted in dark red, gradually transitioning to lighter colours as centrality decreases, with the lowest centrality represented in white.

At Tsim Sha Tsui Station, a majority of signage (40) is connected to three roads, with a substantial portion (18) linked to four roads. Other signages exhibit varying degrees of connectivity, with two signages connected to as many as six roads. Conversely, at Shau Kei Wan station, the majority of signage (11) is connected to three roads, and 10 are linked to four roads. A minority of signage (1 & 6) is connected to two or fewer roads, while the remaining minority (3 & 2) is connected to five or more roads, respectively.

Moving to Charing Cross Station, the analysis reveals a relatively balanced distribution of connectivity among directional signage. A significant portion (30) is linked to three roads. Smaller proportions of signage are connected to fewer or more roads. Similarly, at Albany Park Station, the degree centrality analysis shows a narrow distribution of connectivity among directional signage, with the majority (5) connected to three roads. A notable minority (1) is linked to two roads, and another minority (1) is linked to four roads.

At Britomart Station, the analysis highlights a diverse distribution of connectivity among directional signage, with the majority (15) connected to four roads. Other proportions of signage are connected to fewer or more roads. Finally, at Papatoetoe Station, the analysis indicates a relatively middle level of connectivity among directional signage, with one signage connected to two and three roads, and two signages linked to four roads.

Closeness Centrality

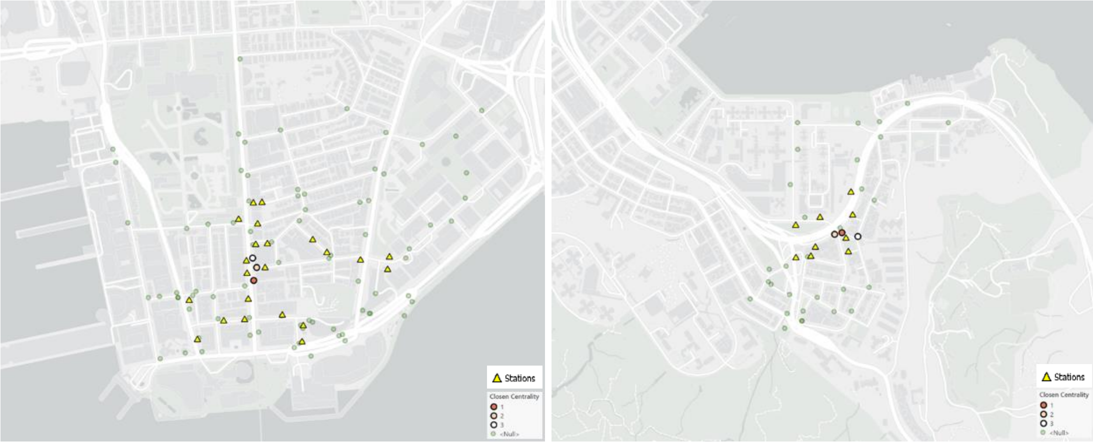


Figure 24: Closeness Centrality Analysis of Train Directional Signs at Tsim Sha Tsui Station (Left) and Shau Kei Wan (Right) in Hong Kong

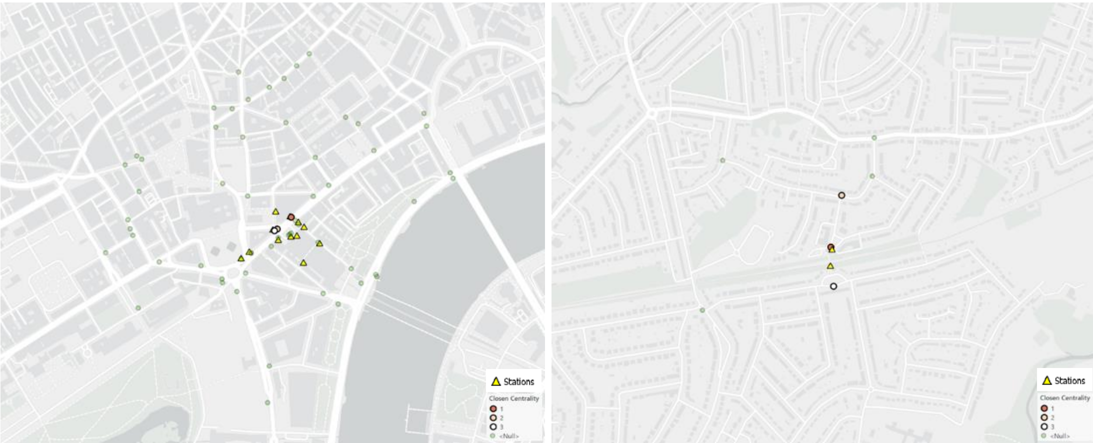


Figure 25: Closeness Centrality Analysis of Train Directional Signs at Charing Cross Station (Left) and Albany Park (Right) in London

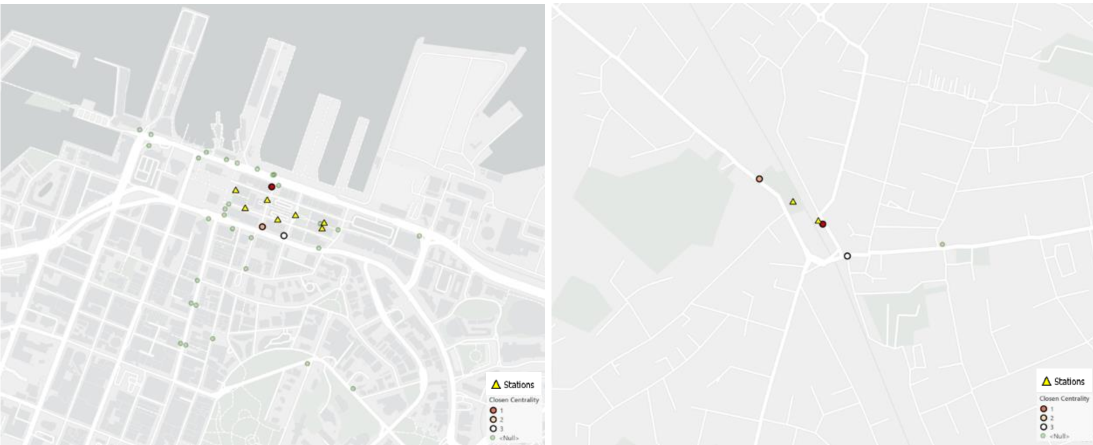


Figure 26: Closeness Centrality Analysis of Train Directional Signs at Britomart Station (Left) and Papatoetoe (Right) in Auckland

Closeness centrality results are displayed on maps from Figure 24 to 26. The highest closeness centralities are red, mid-level are pink, and the lowest one is white. Green symbols represent null values, and station entrances are marked with yellow triangles.

Betweenness Centrality

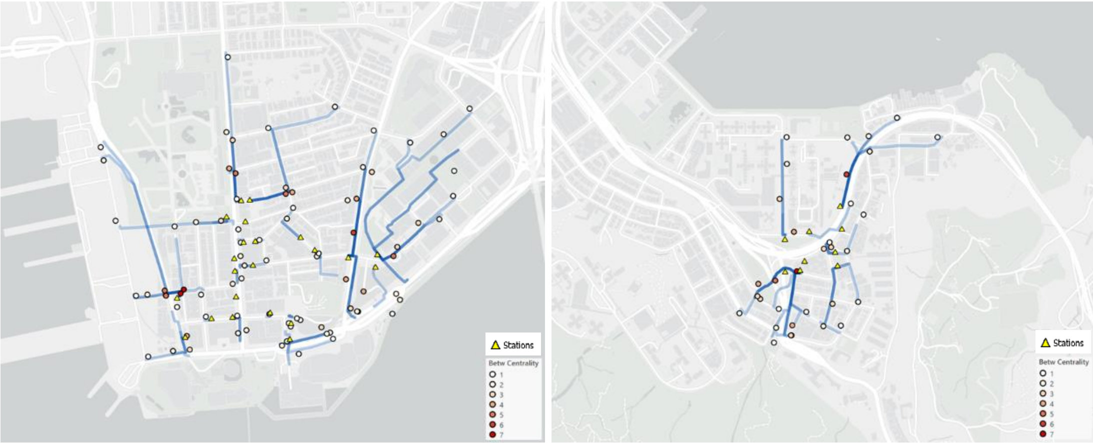


Figure 27: Betweenness Centrality Analysis of Train Directional Signs at Tsim Sha Tsui Station (Left) and Shau Kei Wan (Right) in Hong Kong

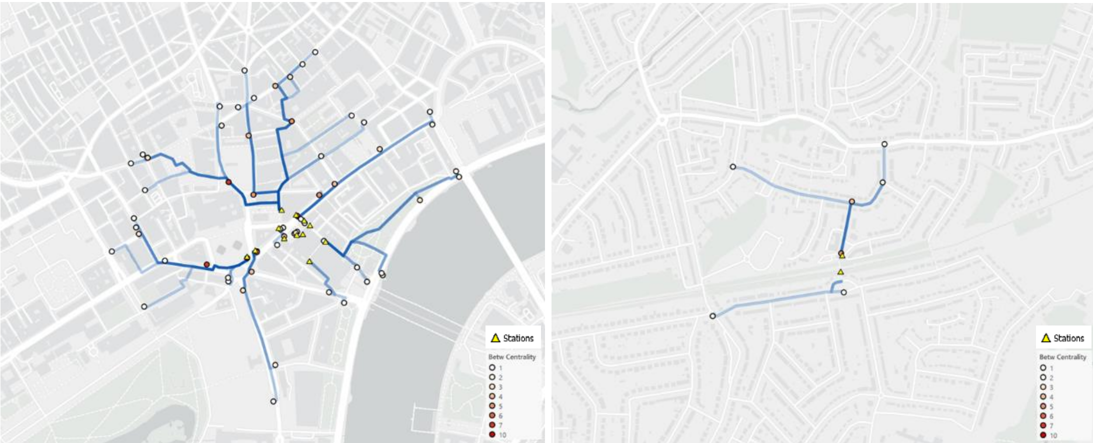


Figure 28: Betweenness Centrality Analysis of Train Directional Signs at Charing Cross Station (Left) and Albany Park (Right) in London

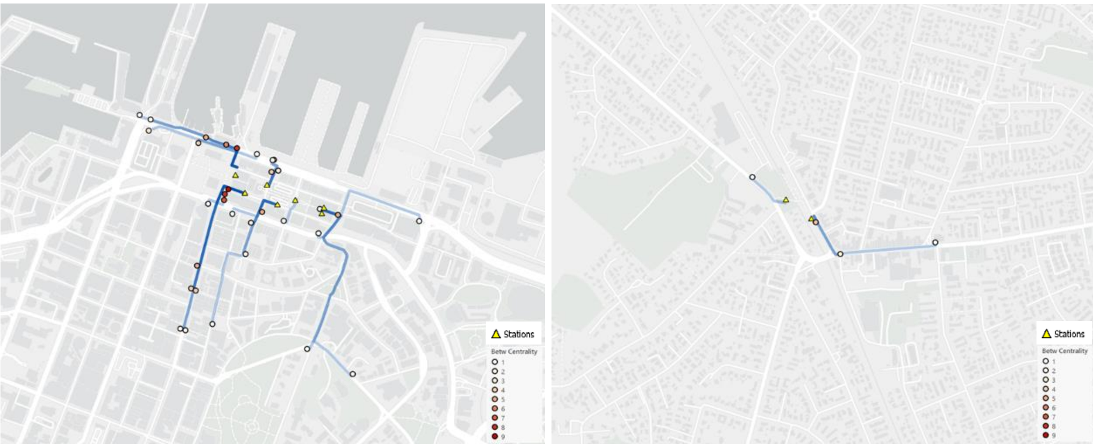
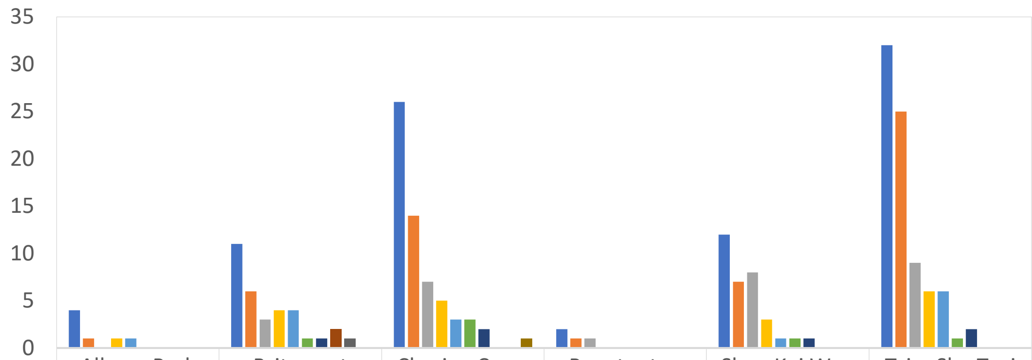


Figure 29: Betweenness Centrality Analysis of Train Directional Signs at Britomart Station (Left) and Papatoetoe (Right) in Auckland



	Albany Park	Britomart	Charing Cross	Papatoetoe	Shau Kei Wan	Tsim Sha Tsui
1	4	11	26	2	12	32
2	1	6	14	1	7	25
3		3	7	1	8	9
4	1	4	5		3	6
5	1	4	3		1	6
6		1	3		1	1
7		1	2		1	2
8		2				
9		1				
10			1			

Table 8: Clustered Column Graph and Table of Betweenness Centrality Analysis Result of Train Directional Signs for Each Study Station

The results of the betweenness centrality analysis are depicted on the maps from Figure 27 to 29. In these visualizations, higher betweenness centrality is represented by darker shades of red, while lower centrality is shown in lighter colours, eventually transitioning to white. Higher betweenness centrality signifies that more shortest paths traverse a single directional signage, whereas lower centrality indicates fewer shortest paths passing through.

At Tsim Sha Tsui station, 32 signages guide passengers with only one shortest path, while 25 signages are intersected by two shortest paths. Nine signages are encountered by three shortest paths, six by four paths, and six by five paths. Additionally, one signage in Tsim Sha Tsui station serves as a crucial junction with seven shortest paths. Shau Kei Wan Station has 12 single-path signages, seven with two paths, and eight with three paths. Three signages facilitate four paths, and one each for five, six, and seven shortest paths.

Charing Cross Station features 26 single shortest path signages, 14 with two paths, and seven with three paths. Five signages facilitate four paths, three with five and six paths, and two with seven paths. Moreover, one signage in Charing Cross Station serves as a crucial junction with ten shortest paths. Albany Park Station has four single-path signages, one with two paths, one with four paths, and one with five paths.

Britomart Station showcases 11 single shortest path signages, six with two paths, three with three paths, and four with four and five paths respectively. One serves as a critical junction with six and seven paths respectively, while another has nine paths. Two signages have eight paths. At Papatoetoe Station, two single shortest path signages guide passengers, one with two paths, and one with three paths.

Discussion

In this study, the spatial dynamics of train station directional signage and its impact on pedestrian accessibility across three diverse cities - Hong Kong, London, and Auckland - as well as within two different dynamic stations within the same city, were investigated. Spatial analysis techniques, including point data analysis and network analysis, were employed to elucidate the relationship between the placement of directional signage and the ease with which individuals can navigate to train station entrances.

Hong Kong

The number of train station directional signages in Hong Kong is 1.68 times and 3.08 times more than London and Auckland, respectively. Similarly, the number of station entrances is 2.21 times and 3.44 times more than London and Auckland, respectively. Merely considering the quantity, without examining the spatial distribution on the maps, the abundance of signages and entrances provides comprehensive guidance. This is because a higher number of signages increases the likelihood of them being positioned around pedestrians, ensuring they can easily spot the guidance from their original position to the station entrance. Furthermore, the ratio of signages to station entrances is comparable at 1:3.67 between the frequently visited Tsim Sha Tsui station and the residential Shau Kei Wan station. This implies that every entrance has at least three signages guiding pedestrians to reach it. In the case of Tsim Sha Tsui, it has the highest number of station entrances and signs compared to all other study stations, although the ratio is not the highest. However, when comparing only the less frequently visited stations, Shau Kei Wan station has four times more station entrances and signages.

Looking at the proportional relationship between train station directional signages and roads across cities like Hong Kong, London, and Auckland, it's evident that directional train station signage varies significantly. In Hong Kong, there's a notably higher proportion of signage compared to other stations, with approximately one signage for every five streets in Tsim Sha Tsui and eight streets in Shau Kei Wan stations. This proportional distribution is strategic, acknowledging that not every street requires signage and considering factors like pedestrian flow and street characteristics. Despite variations, Hong Kong's approach is commendable, reflecting a conscious effort to provide sufficient guidance. Examining the maps reinforces the belief that Hong Kong's directional train station signage is strategically placed. While not evenly spread across every street, signs are predominantly positioned and clustered along major thoroughfares where pedestrian traffic, tourists, and local residents are most likely to pass.

Jumping into the area covered by signs and a box-and-whisker plot of the shortest distance path, the frequently visited station, Tsim Sha Tsui, covered the largest area at 709337.29 m2. However, when comparing two others less frequently visited stations, Albany Park and Papatoetoe, Shau Kei Wan also covered the largest area. Extensive signage around station entrances, as seen in areas like Tsim Sha Tsui and Shau Kei Wan, offers significant advantages. Firstly, it simplifies navigation, reducing commuter stress and time spent locating entrances. Secondly, it serves as prime advertising space (Kellaris & Machleit, 2016), attracting locals to use nearby train station services, thereby raising awareness and encouraging the use of public transportation. Moreover, it enhances the area's aesthetic appeal and identity (Kellaris & Machleit, 2016), particularly in tourist-centric locales. Additionally, it improves urban accessibility and safety by guiding pedestrian flow and promoting sustainable transportation options.

For the shortest distance path from signages to the station entrance in Hong Kong, the range between overall signages and entrance is approximately 0 meters to around 600 meters, and to 420 meters in Tsim Sha Tsui and Shau Kei Wan, respectively. First, this indicates that directional signages mostly likely installed very close to the entrance. Second, the interquartile range (IQR) compared to other stations is less, and upper quartiles fall below 220 meters. Third, both Tsim Sha Tsui and Shau Kei Wan have outliers, with Shau Kei Wan having fewer. Particularly, there are more outliers in Tsim Sha Tsui station that fall outside of the interquartile range. This suggests that the Hong Kong transportation department has focused on installing more signages in the range between 50 meters to 200 meters but has investigated fewer directional signages assets in the further part, concluding that pedestrians have fewer indicators to guide them.

Using Jenks Natural Breaks classification of Kernel Density Estimation in maps, both reveal 3 to 4 prominent clusters. These clusters predominantly occur closer to the station entrances. In Tsim Sha Tsui, two clusters emerge within the shopping mall zone, one adjacent to the park and tourism area, and another along a major road surrounded by entrances. This indicates a centralized approach to outdoor signage placement in densely populated areas, effectively guiding commuters from diverse outdoor vantage points where crowds gather. Additionally, there's a notable trend in the distribution of outdoor directional signs, with a radial pattern emanating from the station's entrance in Tsim Sha Tsui. This radial distribution facilitates guidance for pedestrians approaching the station from multiple outdoor entry points, aligning with the central location of the station in a bustling urban environment. Furthermore, the dispersion pattern of signs in Tsim Sha Tsui shows consistent sign density along streets away from the station entrances. The result of Kernel Density Estimation maps also reveals an increased sign density near major intersections and transport hubs, suggesting a strategic focus on guiding individuals as they approach the stations from various outdoor routes.

On the other hand, in Shau Kei Wan, two clustered areas are situated around grocery stores, shops, markets, and near central bus stations, with another cluster near the entrance of the Shau Kei Wan station. Shau Kei Wan exhibits a dispersed pattern with consistent sign density along less frequented streets, indicating that even in areas with lower foot traffic, signage is present. This suggests that local authorities prioritize not only tourists but also local residents when planning directional signage placement. There is also a notable variation in sign concentration, with both relatively low and high concentrations observed near shops, markets, and central bus stations. Signs are placed strategically to assist individuals in navigating cities more effectively. Directions are indicated near train stations to guide people towards key entry points, intersections, and public transportation nodes. This ensures that individuals can easily determine the direction to their desired destination.

The result of the Average Nearest Neighbour analysis reveals that the Nearest Neighbour Ratio (NNR) in Tsim Sha Tsui is 0.82. This indicates that directional signages in Tsim Sha Tsui tend to cluster together (Grekousis, 2020), as the observed average distance between them is less than the expected average distance. Ripley's K-function chart further confirms this observation, as the observed K values consistently exceed the higher confidence envelope along the expected value (Grekousis, 2020). This statistically significant clustering occurs at both smaller and further distances, suggesting a pronounced clustering effect throughout the area. This clustering phenomenon is particularly notable in certain areas of Tsim Sha Tsui, such as shopping malls, leisure zones, and along major streets where a variety of shops are located.

Given the high tourist attraction in these areas, optimizing the arrangement of signages can enhance tourism management by efficiently guiding tourists to station entrances, thereby improving accessibility in crowded areas. However, it is important to ensure adequate signage coverage in remaining areas where clustering is not as pronounced. This ensures that all areas receive appropriate guidance and support, enhancing overall navigation and convenience for residents and visitors alike. In Hong Kong, this aspect has been effectively managed.

On the other hand, when examining Shau Kei Wan, the NNR is 1.12, indicating a random spatial pattern (Grekousis, 2020). This is also reflected in the chart of Ripley's K-function, where the observed k values are close to the expected k values and fall within the upper and lower confidence envelope. This suggests a mix of clustered and dispersed directional signages in the area (Grekousis, 2020). The dispersed pattern in residential areas can be attributed to residents' familiarity with the area, as they navigate to station entrances in their daily lives. Conversely, the clustered pattern observed along major streets, featuring shops, restaurants, street markets, and central bus stations, suggests a deliberate concentration of signages in areas frequented by diverse groups of people. Increasing the number of clustered signages in these areas can effectively guide individuals with varying purposes without causing confusion in wayfinding. From this perspective, the strategic placement of directional signages appears to target specific types of pedestrians, aligning with the varied needs and navigation behaviours of different demographic groups within the area.

Now, analysing the accessibility of directional signages from a network analysis perspective using centrality measurements, let's focus on degree centrality. A higher degree centrality indicates that more directional signages are effectively utilized at intersections (Golbeck, 2015), thereby reducing the possibility of individuals being unable to find their way to the station entrance. With reference to the clustered column graph and table of degree centrality analysis, it is notable that in Tsim Sha Tsui station, the results for three (grey) and four (yellow) degree centrality are significantly higher than those of all other studied stations in different cities. This suggests two key points: firstly, there is a greater number of signages installed, and secondly, more than 70% of these signages are located at intersections and junctions with multiple roads, allowing for more efficient directional guidance. However, it is worth noting a contradiction: some intersections with more than three or four roads connected to them have multiple instances of signage. While this may not be the most cost-effective approach, as it does not reduce the number of signages at the same intersection, it offers thorough guidance for pedestrians approaching these intersections from different directions. Additionally, it addresses potential blind spots on the street that may arise if only one signage were provided at each intersection. Further examination of the map reveals that intersections with a degree centrality of one or two are likely situated at dead-end roads near parks and popular streets. This intentional placement reinforces the notion that signages are strategically positioned to cater to specific pedestrian needs, even in less frequented areas. At Shau Kei Wan station, although the number of signages is not as extensive as at Tsim Sha Tsui station, the results indicate that nearly 80% of signages are positioned at more than two intersections and junctions. This suggests a more efficient utilization of directional signages, particularly in busy areas such as shops, restaurants, and street markets. Conversely, signages with one or two degree centrality are typically situated near residential areas. This is attributed to the fewer intersections in residential areas, resulting in a lower density of directional signages.

For the benefit of higher closeness centrality, directional signages can efficiently guide individuals to all station entrances, ensuring quick access. This optimized pedestrian flow is particularly beneficial in areas with a high volume of people, as it helps manage and streamline the movement of pedestrians entering the station (Saxena & Iyengar, 2020), thereby reducing congestion and potential bottlenecks. Moreover, in situations where a large crowd is seeking station entrances, such as during peak commuting hours or special events, directional signages with high closeness centrality contribute to public safety by preventing overcrowding. In Tsim Sha Tsui station, the concentration of high closeness centrality is notably evident along the popular street, Nathan Road, which is lined with hotels, shops, famous shopping malls, churches, and commercial buildings. This strategic placement indicates that directional signages with high closeness centrality are well-positioned to serve the diverse needs of visitors and residents alike. In Shau Kei Wan station, while high closeness centrality may also contribute to population diffusion, particularly around popular areas such as the main street East and street markets, it may not be as effective as in Tsim Sha Tsui due to lower visitor traffic.

In terms of betweenness centrality, while each directional signage holds equal importance and plays the same role, those with higher betweenness centrality have a greater impact when they are absent or damaged (Saxena & Iyengar, 2020). However, this is not always the case because not every directional signage lies on the shortest path between other directional signages. Some directional signages with a betweenness centrality of one may stand alone, being the only signage on their respective shortest path. Typically, directional signages with higher betweenness centrality are positioned closer to the station entrance, as they guide pedestrians along the shortest path to the station entrances. Furthermore, ideally, directional signs at train stations should offer the shortest route for guiding passengers to entrances. It is anticipated that a sequence of signs should be connected along this shortest route. Thus, a higher betweenness indicates a sufficient number of train station signs provided and installed along the shortest path. In Tsim Sha Tsui station, around 70% of directional signages have one or two shortest paths passing through them, indicating that they are located either at the furthest distance from the stations or close to the central area of Tsim Sha Tsui. The remaining 30% are relatively close to one of the station entrances. Additionally, there are 15 routes that overlap with other shortest paths, suggesting a well-proportioned distribution that facilitates accessibility through a well-organized guiding system. In comparison, Shau Kei Wan station boasts more than 40% of directional signages along the shortest paths and also has 7 routes that overlap with other shortest paths, along with fewer standalone signages. Although this is only half of what Tsim Sha Tsui station offers, it still provides similar benefits in enhancing accessibility. While a larger number of high betweenness centrality points are not expected, with 30% in Tsim Sha Tsui and 40% in Shau Kei Wan, it is a remarkable percentage and is evident in guiding passengers effectively in Hong Kong. It's even surprising that Shau Kei Wan, being a station with less frequent services and predominantly residential, achieved such results which underscores its effectiveness.

London

When comparing two stations, Charing Cross station and Albany Park station, solely by examining their maps, a significant contrast in the quantity of train station signages between these two London stations becomes evident. The same holds true for the number of station entrances. In Charing Cross station, there are more than eight times the number of signs compared to Albany Park station, and six times the number of station entrances. This directly reflects the heavy focus of the London local authority on representative locations in London that are frequented by tourists and locals. While this emphasis is beneficial, the lack of signs in stations frequented by residents may lead to confusion for those unfamiliar with such stations (Zheng et al., 2007), as well as residents who may become lost without clear signage, thus prolonging their walks on the streets. Compounding this issue is the fact that, unlike in Hong Kong, where pedestrians are common, in London, people are more likely to drive than walk. This means that if you get lost, there may not be readily available assistance from pedestrians, leaving you to navigate and seek help on your own. Interestingly, Charing Cross station boasts the highest ratio of signs to station entrances, with five signs per station, offering comprehensive navigation. This targeted approach is believed to cater well to tourists visiting the renowned Charing Cross station.

Within a 500-meter radius of Charing Cross station, there is one directional signage for every 6 roads, whereas in Albany Park station, there is one signage for every 20 roads. Once again, a significant contrast is evident when comparing stations within the same city of London or when comparing to stations in Hong Kong. While Charing Cross station exhibits a similar proportion between signages and roads as seen in Hong Kong, Albany Park station demonstrates a notably lower number of signs available for pedestrians. It's important to note that signage tends to be clustered near the station entrance, meaning there are no signs available for pedestrians walking from regions outside the station's vicinity. In the outer areas, individuals may need to traverse more than 20 streets before encountering a sign to guide them to their destination, which is unfavourable for pedestrian navigation. Although this statement is solely based on numbers, it may not be entirely accurate, as pedestrians typically walk directly to the station entrance along the shortest path, without necessarily traversing every road before reaching their destination.

With a coverage area of 638195.6 square meters of signage in Charing Cross station, it has performed admirably by extending its reach to various locations such as squares, galleries, numerous prestigious shopping malls, theatres, museums, hotels, countless restaurants, and hospitals, among others. The station serves as a hub for diverse activities, benefiting from the guidance provided by its extensive signage network, which aids individuals attending different events in finding their way to the station entrance. Furthermore, it is noteworthy that the coverage area of signs in Charing Cross station even overlaps with three other nearby stations: Embankment station, Covent Garden station, and Leicester Square station. In contrast, the coverage area of signage in Albany Park station is considerably smaller and primarily caters to local residential housing and schools. This limited coverage area is reasonable given that residents and students may typically drive to the station from their homes or are familiar with the route to the entrance without relying on signage.

Both Charing Cross and Albany Park stations exhibit a balanced distribution of signs along their respective shortest paths, with signs placed both close to the station entrance and further away. While there is a slightly higher number of signs in Albany Park station placed further from the entrances. It is intriguing that despite the smaller coverage area of signs in Albany

Park, the longest shortest path is remarkably close to that of Charing Cross station. This implies that pedestrians traveling from greater distances to the entrance in Albany Park can benefit from signs placed at a distance. Returning to Charing Cross station, it is commendable that an equal number of signs are present in both short and long-distance areas from the station entrance. This demonstrates the thorough consideration by local authorities in planning and installing these signs, a practice that likely extends to other stations in London. This consistent approach aligns with the norm across London's studied stations, indicating a comprehensive signage system designed to guide both local residents and tourists consistently along the shortest route to the station entrances.

As the number of signages directly influences the colour intensity and relative size of spots in KDE maps (Grekousis, 2020), this KDE analysis serves the purpose of identifying specific clustering patterns across the studied stations in London. It is evident from the KDE results that there is a prominent cluster area surrounding the station entrance. This result reflects and aligns with the underlying purpose, which is to ensure that pedestrians seeking to access Charing Cross station can easily locate its entrance. Charing Cross station is an underground station that is built inside a building, with its main entrance integrated into the building structure. Without a sufficient number of tall signages displaying the London transportation logo, individuals unfamiliar with the area may overlook the fact that the station entrance is concealed within the building. Concluding, these signages are strategically placed to effectively guide pedestrians and fulfil their intended purpose. Moreover, relatively low-intensity clustered areas are observed across Charing Cross station in the KDE maps, displaying a relatively even distribution of signs throughout the station vicinity. As mentioned in the results, there is an intriguing series of points, forming almost a line along the streets in the northern part of Charing Cross station. This phenomenon is attributed to the presence of various prominent shopping malls and department stores in that area. This further underscores that signs are purposefully placed along streets where many people would frequent or pass by, rather than being randomly distributed. In Albany Park, although the maps display four prominent cluster areas, an entrance-centralized approach is evident as signages are predominantly placed around the station entrance. Notably, no signages are found on the east side and further south side of the entrance when zooming out to view the results. However, clustered areas are still noticeable due to the small amount of signages placed close together, with one prominent cluster surrounding the station entrance which is beneficial for guiding pedestrians.

The results of the Average Nearest Neighbour analysis indicate clustering at Charing Cross Station with a Nearest Neighbour Ratio (NNR) of 0.86, while at Albany Park station, it's 2, indicating dispersion (Grekousis, 2020). This disparity is understandable given that signage at Charing Cross station is intentionally clustered along major streets, catering to specific pedestrian demographics such as shoppers, tourists, relaxation seekers, and food enthusiasts. Supporting these findings, Ripley's K-function analysis reveals a higher observed K compared to expected K in the initial distance, indicating clustering, but a subsequent drop below expected K, suggesting dispersion at greater distances (Grekousis, 2020). This shift implies a diminishing clustering effect as signage becomes less concentrated with distance. This phenomenon can be attributed to the strategic clustering placement of signs near station entrances, which in turn causes dispersion at farther distances.

On the other hand, the dispersed spatial pattern observed at Albany Park station aligns with the characteristics of this particular station. Firstly, the station serves a diverse community encompassing residents, students, and others familiar with the area, thus reducing the necessity for extensive signage. Residents and frequent commuters are likely accustomed to the station's location, diminishing the need for prominent signage directing them. Secondly, unlike Charing Cross station where specific streets require concentrated signage due to high pedestrian traffic, Albany Park lacks such high-traffic areas, reducing the demand for extensive and clustered signage placement. While major intersections may receive some spatial weight, leading to an exhibition of clustering, it is not as pronounced compared to Charing Cross stations. Additionally, residential stations like Albany Park typically experience lower foot traffic compared to urban hubs like Charing Cross. With fewer pedestrians relying on foot traffic and a greater reliance on vehicular transportation (GLA, 2014), the need for visible signage diminishes further. Consequently, there is less emphasis on clustering signage around the station itself, contributing to the dispersed spatial pattern observed. Examining the signage placement on maps at Albany Park station reveals a strategic approach focused on key intersections and an extensive dispersed distribution throughout the area. Signage primarily directs pedestrians to turn right or left, it does not instruct them to 'keep walking forward'. Following this rule, signage is not found even at other key major road intersections, thus leading to the dispersion pattern observed in train station direction signage.

From a network analysis perspective, the degree centrality results in London Charing Cross station reveal that 28% of the signages are connected to one or two roads, while the remaining 72% are connected to more than two roads. This high percentage of signages with high degree centrality is believed to offer several benefits. Firstly, it facilitates navigation for tourists who often encounter confusion or get lost at intersections. Directional signages can redirect them to their destinations efficiently, enhancing their overall experience in the area. Additionally, the prevalence of directional signages at intersections and junctions contributes to cost and time effectiveness. In terms of cost, it eliminates the need for installing signages on every single street, focusing instead on key and useful locations where guidance is most needed. This targeted approach minimizes expenditure while still ensuring effective navigation for pedestrians. In terms of time, directional signages save people the effort of wandering around streets in search of directions, allowing them to reach their destinations more efficiently. By providing clear and concise guidance at crucial points, these signages streamline the navigation process, ultimately saving time for pedestrians. Moreover, installing directional signages at these critical points helps alleviate the high density of pedestrian traffic exiting from prominent landmarks such as shopping malls, department stores, theatres, or squares near the station. Without proper guidance, such congestion could escalate into potentially hazardous situations like overcrowding and stampedes. Hence, the presence of directional signages plays a crucial role in maintaining public safety in crowded urban environments. Despite constituting only 28% of the total, signages with lower degree centrality still serve a critical function. In areas where the train station entrance may be obscured from view, these signages play a vital role in guiding pedestrians to their desired destinations. Their strategic placement ensures that even in areas with fewer road connections, pedestrians can navigate effectively, enhancing accessibility and reducing instances of disorientation. At Albany Park station, none of the signages are positioned at dead ends, demonstrating the effective utilization of signage by the London local authority at intersection locations.

The high closeness centrality of signages in both Charing Cross Station and Albany Park in London has not delivered the anticipated impact. In Charing Cross, the high closeness centrality of signages does not seem to be significantly influenced by any specific factors. There are no apparent reasons for sudden influxes of crowds or congestion at these locations. Even if there were, the usefulness of these high closeness centrality signages would be limited, as people can already see the station entrance from these positions. However, it is believed that they may still provide some assistance, albeit in smaller quantities. Similarly, at Albany Park station, the situation is presumed to be similar. Despite the high closeness centrality of signages, there are no clear contributing factors to explain their prominence. The absence of specific reasons for increased pedestrian traffic or congestion diminishes the potential effectiveness of these signages with high closeness centrality.

Charing Cross station stands out as the only station boasting signages with a betweenness centrality rating of 10, the highest among all stations. This signifies that 10 of the shortest path routes pass through these particular directional signages. Upon examining the maps, it becomes apparent that these shortest paths converge, contributing to this station's highest betweenness centrality. This paints a picture of a robust navigation system. Not only does it save costs by not requiring signage in less frequented areas, but it also efficiently guides individuals along a single, optimal route. When pedestrians observe the directional flow towards the station entrance indicated by these signages, they naturally follow suit, streamlining the navigation process. With over 40% of the signage possessing a betweenness centrality of just one, it's important to note that even though they have the lowest centrality, they still offer advantages. These signages play a crucial role in guiding individuals from distant points to closer proximity and ultimately to the station entrance. This distribution represents a great proportion between high and low betweenness centralities. Contrasting this with Hong Kong's Tsim Sha Tsui station, it's evident that the shortest path routes in Charing Cross station, demonstrate a station entrance centralized approach. It's easy to imagine crowds naturally gravitating towards the station entrance in a centralized manner. Charing Cross station may not facilitate such smooth pedestrian flow towards its entrance, primarily because people tend to cluster together when entering, a challenge less prevalent in Tsim Sha Tsui station. This phenomenon can be attributed to the spatial clustering distribution of station entrances in Charing Cross station. In Albany Park, despite the relatively lower quantity of signages, the betweenness centrality map reveals a converging pattern. This indicates that even with fewer signages, they efficiently guide pedestrians from significant distances to the station entrance through a network of connected shortest path routes.

Auckland

Finally, in the examination of directional signages installed at Britomart and Papatoetoe stations in Auckland, a notable contrast emerges akin to that observed in London. Britomart, a frequently visited station, boasts a significantly higher quantity of directional signages compared to Papatoetoe, which serves a less frequented and residential area. At Britomart, each station entrance averages 4.71 signages guiding pedestrians and passengers— the number surpassing even the signage-to-entrance ratio observed in Hong Kong. The ratio of signages to roads in Auckland, particularly at Papatoetoe, reveals a different picture, with a mere 0.03 signages per road. Britomart's ratio isn't significantly higher than Papatoetoe's, standing at only 0.08 signages per road. While the relatively high signage-to-entrance ratio at Britomart suggests a commendable investment by the Auckland local government to enhance visitor navigation, the low quantity of signages and minimal ratio of signs to roads in both Britomart and Papatoetoe indicate room for improvement. This suggestion stems from the reality that individuals often traverse considerable distances before reaching the station entrance, increasing the likelihood of losing direction along the way. This challenge is exacerbated in Auckland due to the comparatively longer distances between stations compared to cities like Hong Kong and London. Moreover, increasing the number of signages would aid pedestrians in spotting directional indicators more easily, particularly in spacious environments with fewer passersby. In Papatoetoe, where only two signages adorn each station entrance—the lowest among all studied stations, with a mere four signages in total—the need for improvement is evident. While this scarcity of signages is somewhat understandable given that the area primarily serves residents, the influx of foreigners, such as international students and workers, necessitates enhanced navigational aids. Many of these newcomers rely on Papatoetoe as a departure point for reaching central Auckland, specifically Britomart, and are unfamiliar with its station entrance locations at Papatoetoe.

Considering the first two parameters, it's unsurprising that the average coverage area by signages in Papatoetoe is six times smaller than that in Hong Kong's Shau Kei Wan. Furthermore, analysis of the distribution of signages within a 500-meter buffer zone around Britomart and Papatoetoe stations reveals a concentration of signages near the stations themselves. This contrasts with Hong Kong and Shau Kei Wan, where signages are more evenly distributed across greater distances. This suggests that individuals may only encounter directional signages when they are already close to the station, rendering them potentially ineffective. Therefore, increasing the number of signages from shorter to longer distances would expand the coverage area and enhance navigational assistance. However, it's worth noting that the local government likely considered factors such as distance from the station when planning signage placement, as evidenced by the fewer signages observed in areas farther from the station.

The Kernel Density Estimation map illustrates clusters of relatively high signage densities near the northwest side of Britomart station's entrance, with the majority of remaining signages concentrated along major streets like Queen Street. Consequently, large areas surrounding the east and south of Queen Street, as well as the west side, appear devoid of train station direction signages, highlighting notable gaps in signage coverage. On the north side, marked by the elongated rectangular shape on the map, there are various amenities including the Cruise Ship Terminal, ferry terminal, freight terminal, as well as leisure spots like the lighthouse and waterfront area. Additionally, there are hotels, bars, and restaurants in abundance. However, Surprisingly, data shows a lack of directional signages in this bustling area, posing a concern for tourists and visitors, particularly those needing to transfer from the ferry terminal to the train station. However, a paradox arises in this setting as the direct path from the ferry terminal to Britomart station entrance might alleviate some confusion, albeit the need for clear signage remains. Directional signages predominantly line Queen Street, which hosts numerous shops, cafes, gaming venues, cinemas, and eateries. While this signifies a commendable effort in facilitating navigation for people from various distances and directions, areas lacking signage, particularly near the universities and business hubs, necessitate additional directional sign installations. Comprehensive navigation aids are crucial, especially for international students and workers commuting to and from these locations. At Papatoetoe station, with only four signages installed, the density level appears to be uniform along major roads. However, given the presence of a leisure centre and shopping market nearby, more signage is recommended. Even in residential areas where commuters may need to navigate through intersections and turns to reach the station, increased signage can greatly enhance convenience and efficiency, akin to practices observed in Hong Kong and London.

After discussing the results of Kernel Density Estimation mapping, it's not surprising to find that Britomart shows clustering with an NNR (Average Nearest Neighbour Ratio) of 0.8285, while Papatoetoe exhibits dispersion with an NNR of 2.7841 (Grekousis, 2020). The map reveals clustered spots around Britomart station, particularly near the station entrance where the shopping mall is located, as well as along major central streets bustling with clothing shops, cafes, gaming places, cinemas, and various eateries and restaurants, as mentioned earlier. This suggests that the local government has strategically positioned signage in these areas. However, it's essential to also consider installing signage in other areas to ensure comprehensive coverage. In contrast, Papatoetoe displays a dispersed pattern, likely because signage is placed at evenly distributed intersections throughout the area. Although Ripley's K-function shows similar results with observed K values consistently higher than expected K values, the fact that they remain below the upper confidence envelope suggests that the clustering is not statistically significant (Grekousis, 2020). This indicates that there may be some noise or random factors affecting the distribution, which should be investigated further in subsequent studies.

In terms of degree centrality, Britomart stands out from Charing Cross and Tsim Sha Tsui. While both Charing Cross and Tsim Sha Tsui exhibit a degree centrality of three (indicated by the grey bars), Britomart surpasses them with a degree centrality of four, represented by the tallest yellow bar. This suggests a higher concentration of signages positioned at the central intersections of four roads in Britomart (Golbeck, 2015), which are likely key areas of pedestrian traffic. Conversely, the presence of only one degree centrality (depicted by the blue bar) at Britomart indicates a relatively lower number of signages positioned at road dead ends. This discrepancy may stem from differences in street layouts, particularly notable in Auckland. Furthermore, Britomart shows only two signages connected to five roads, indicating the unique street configurations in Auckland that may limit such placements. Conversely, Papatoetoe, despite having fewer signages overall, maintains a generally normal degree centrality, ranging from connections with two to four roads. Both stations, however, demonstrate a satisfactory level of signage distribution, ensuring pedestrian navigation within the normal range of degree centrality.

The highest closeness centrality in Britomart holds significant utility for several reasons. Situated near a bustling harbour area, Britomart boasts a variety of facilities, including the cruise ship Terminal, ferry terminal, freight terminal, and leisure amenities like the lighthouse and waterfront area. With a considerable influx of passengers disembarking at Britomart Pier and transferring to the train station for onward journeys, the top closeness centrality node positioned near the pier and the train station serves as a pivotal point for efficiently dispersing crowds during peak times (Saxena & Iyengar, 2020). Moreover, the second and third highest closeness centrality nodes play a similar vital role. Located along routes leading from the other side of harbour area, these nodes cater to individuals traversing from the adjacent streets filled with diverse shops and large international businesses, as well as universities. This connectivity ensures efficient movement towards the station entrance (Saxena & Iyengar, 2020), facilitating smooth travel from south to north. Conversely, in Papatoetoe, the analysis yields contrasting results. Despite identifying the top three closeness centrality nodes, the index of closeness centrality may not offer as much utility. The surrounding environment likely lacks the concentrated foot traffic and diverse amenities found near Papatoetoe, diminishing the necessity for high closeness centrality directional signages for crowd dispersal and transit facilitation.

Britomart exhibits a similar pattern to Charing Cross, where instead of scattered shortest path routes like in Hong Kong, the shortest routes in the betweenness centrality maps display a clear, tidy, and well-converged path starting from further distances and connected to the signages along a series of connected shortest path routes. This pleasing result is further supported by the table and column graph, showing the number of betweenness centrality ranging from nine to one, with nearly 50% of the signages being converged into common shortest routes. This pattern proves useful as pedestrians won't get confused and can simply follow the signage one by one to reach the station entrance. Furthermore, it indicates that the signages were thoughtfully planned by the local government rather than randomly, which is commendable and worth referencing. Surprisingly, despite other parameters not performing as well as in Papatoetoe, betweenness centrality actually shows excellent results as it mirrors the outcome observed in Britomart.

Conclusion

Here, conclusions drawn from findings uncovered through our research will be discussed. Firstly, the number of train station directional signages has been identified as an important parameter affecting accessibility. While a greater number of signages can facilitate navigation and expose more people to them from their departing position, an excessive amount can lead to confusion when pedestrians are overwhelmed by the multitude of guiding services provided by the signages. Therefore, it is suggested to consider the number of entrances and streets when determining the appropriate quantity of signages based on a predetermined ratio. Across six different stations, such as Tsim Sha Tsui and Shau Kei Wan in Hong Kong, as well as Charing Cross and Britomart in London and Auckland respectively, a successful balance has been achieved. It can be concluded that a ratio higher than 1:3.6, indicating that each station entrance should ideally have 3.6 or more signages, is effective. Furthermore, stations like Albany Park and Papatoetoe, primarily serving residential areas with limited amenities, may benefit from an even higher ratio of signages to station entrances due to the expansive nature of their station areas. However, it's important to note that the effectiveness of signages is not solely determined by their quantity, but also by their placement. Placing numerous signages on every street without considering their effectiveness in guiding pedestrians to the station entrances would be counterproductive, wasting both time and resources. Therefore, careful consideration of signage placement is essential to ensure they serve their intended purpose effectively.

The ratio of roads to signages might not be the most suitable parameter to consider for several reasons. Firstly, the number of roads can vary significantly based on the city's road structure and layout pattern. If more roads automatically imply more signs, it could lead to confusion. Therefore, the criteria for determining the number of signages consider to be based on factors other than simply the number of roads. Secondly, different cities may have many minor roads that are rarely crossed, while having fewer major roads. In such cases, relying solely on the ratio of roads to signages for signage placement may not be effective. While the ratio of roads to signages can still be considered as one of the parameters, its weight should not be as significant as the ratio of station entrances to signages. It's essential to take into account various factors when deciding the appropriate number and placement of signages to ensure effective navigation without causing confusion.

This study concludes that coverage area is a crucial parameter for determining the placement of signage, directly related to the furthest travel distance and longest travel time. Among the stations studied, Tsim Sha Tsui and Charing Cross exhibit the largest coverage areas. In dense cities like Hong Kong and London, it's advisable for signage coverage areas to be as large as possible, while ensuring they do not overlap with the coverage areas of adjacent stations. This suggests that the boundary of the coverage area should ideally lie midway between two adjacent stations. However, in cities like Auckland, where pedestrians may not feasibly walk between stations due to long distances, a range of 500 meters to 700 meters is recommended. This range is suggested because a 500-meter radius may not adequately cover the station's surrounding community areas. Moreover, areas such as shopping malls and major streets with restaurants, cafes, cinemas, etc., can extend beyond a 500-meter radius. Conversely, a coverage area cannot be too small. For instance, in Papatoetoe, the coverage area is six times smaller than that of Hong Kong's Shau Kei Wan. In such cases, individuals may only receive navigation assistance when they are very close to the station, possibly even being able to see the station before spotting the directional signage. This renders the directional signage potentially useless.

For ensuring the overall shortest distance and time from each signage to its closest station entrance, it's recommended to strive for a balanced distribution, aiming for a normal distribution of signages in terms of both distance and time. This ensures that regardless of the distance, adequate signage is present to guide individuals effectively. Charing Cross station serves as an excellent example, showcasing a normal distribution of shortest distances and times for its station signages. This means that the majority of signages should not be placed solely at either the farthest or closest points to the station entrances. Placing all signages extremely close to the station entrance may not necessarily aid pedestrians. If signages are positioned too close, pedestrians might spot the station before noticing the directional signages, rendering them ineffective. However, there are exceptions. For instance, if the station entrance is concealed within a building, accessible only underground, or tucked away sideways, then close proximity of signages can indeed be beneficial for navigation.

Using Kernel Density Estimation analysis, the clustered areas around train stations can be observed on maps. Typically, locations such as shopping malls, transportation hubs, major streets with various shops, eateries, cinemas, activities, and sports centres are expected and suggested to have denser spots. While, with the exception of Albany Park and Papatoetoe, every station has met this criterion, there is still room for improvement as some areas may have been overlooked. It's also worth noting that in certain areas, signage clusters don't necessarily need to be in an aggregated form. Instead, a series of aligned signages can be effective, as seen in Tsim Sha Tsui, Shau Kei Wan, Charing Cross, and Britomart. Therefore, this lined-up clustered format is recommended for such scenarios.

Observing the near average ratio reveals an intriguing phenomenon: frequently visited stations tend to display clustered patterns of directional signages, irrespective of the city, whereas stations in residential areas exhibit dispersed patterns. This discrepancy highlights the varying requirements of these locales. In frequently visited areas like commercial districts, tourist spots, or major transportation hubs, there's a heightened demand for clear navigation due to the influx of unfamiliar visitors. To efficiently guide these individuals, local authorities are advised to install more directional signage, creating clustered patterns. Conversely, residential areas experience less foot traffic and fewer visitors compared to commercial zones. Residents are typically familiar with their neighbourhood’s layout and require less directional assistance, resulting in fewer signage installations by local authorities. Moreover, certain streets may necessitate a clustered signage pattern due to blind spots, where visibility is obstructed. In such cases, additional signage is crucial for both safety and navigation. Multiple signages at intersections might be necessary to effectively address these blind spots and ensure clear guidance for pedestrians. In conclusion, frequently visited stations are advised to adopt a clustered or random pattern of directional signages to efficiently guide the influx of visitors. Conversely, stations in residential areas, with fewer visitors, are recommended to employ a dispersed or random pattern of directional signages, catering to the needs of the local residents.

In Ripley's K-function results, it's not anticipated for the observed K to consistently exceed the expected K for every station across all distances, indicating a clustered pattern. This discrepancy arises from specific areas demonstrating clustering at directional distances, while in regions with lower foot traffic, the observed K values may fall below the expected K. Hence, it's essential to consider the environmental context of each station when planning to achieve more effective directional signage.

Degree centrality is a valuable tool for assessing outcomes. While a higher number of high degree centrality nodes indicates effective placement of directional signage at intersections (Golbeck, 2015), which is cost-effective and enhances clarity of junction positions, a lower number of low degree centrality nodes should not be seen as a disadvantage. On the contrary, it can be beneficial. For instance, consider a scenario where a station entrance is located far from the end of a major street, making it invisible from start point. In such cases, signage installed midway along the street, representing a two-degree centrality, would be recommended. This principle can also be applied to areas with existing blind spots around station entrances, where signage indicating their presence would improve visibility. Therefore, a low degree centrality does not necessarily imply inefficiency. Intersections with numerous connected roads should be reviewed for signage installation, and a balanced distribution of high and low signages on roads is advisable.

Closeness centrality presents a paradoxical nature. On one hand, it measures the average closest distance or shortest time to access all station entrances, proving beneficial in locations with significant population influx, such as grand activity venues or shopping malls, where efficient navigation is crucial. However, the paradox arises when only a few or even only one person needs to access these points. They only require the nearest entrance, rendering the high closeness centrality less useful in quantifying signage effectiveness. Moreover, irrespective of high or low closeness centrality, directional signages typically display only one arrow, limiting their ability to guide large population influx in multiple directions with a single sign. The highest closeness centrality is inherently located at the central point of all entrances, which may not necessarily coincide with these high-traffic areas. Additionally, remaining signages still serve their intended function. While high closeness centrality is advantageous near locations with large population inflows, it should not necessarily be interpreted as a negative outcome elsewhere.

Betweenness centrality yields dual insights. Firstly, it assesses the number of shortest path routes passing through each signage. A higher count indicates that the signage lies along more converged shortest paths. Secondly, examining the distribution of these shortest paths on the map reveals whether they are scattered or neatly converge along a series of connected signages. Hence, it is highly recommended for such assessments. All study stations exhibit a positive result in terms of betweenness centrality, irrespective of their result quality or coverage area parameters. Both high and low betweenness centrality values suggest positive outcomes. However, a trend favouring slightly higher betweenness centrality is preferred, as it indicates more converged shortest paths, thereby enhancing signage effectiveness and reducing travel time on incorrect routes. In conclusion, it is ideal for the furthest signages to connect with mid-distance signages as well as the shortest distance signages, thereby creating a continuous sequence of shortest paths leading to the station entrance. This integrated network ensures that pedestrians have a clear and efficient route to follow. By strategically placing signages in this manner, navigation can be optimized, ultimately enhancing the overall accessibility of the station.

Summary

Regarding the highlighted findings, every station in each city has done a commendable job. Directional signages are strategically positioned and well-designed, significantly enhancing pedestrian accessibility to train stations. Signage in each studied city's station has its merits. Our findings have demonstrated that directional signage significantly aids pedestrian access. However, there is still room for further enhancements in the signage placed within each station to strive for perfection.

Since each station is influenced by different factors affecting the spatial placement and distribution of signages, it is challenging to judge from a singular perspective. As a result, there is no standard answer to whether a navigation system of direction signages is perfect or subpar. Instead, the signages in each station area need to be assessed on a case-by-case basis. Ultimately, as long as directional signage successfully and effectively guides people to the entrance of a train station, it fulfils its fundamental intended purpose.

Limitations

Limitations exist within our project due to lack of prior research on this topic, resulting in the absence of definitive standards or criteria to reference.

For data collection methods, the point data source for directional signage in Hong Kong may not be regularly updated through the government website, potentially leading to inaccurate signage counts. Additionally, data collection involved researchers walking the streets of Tsim Sha Tsui and Shau Kei Wan to identify signs, leaving room for potential oversight. Regarding the point data of signages in London and Auckland, it relies solely on Google Street View in Google Maps. Consequently, certain signages may be situated in blind spots or obstructed by construction, and road accessibility for Google's vehicles may vary. Moreover, the timing of Google's vehicles captures varies, with some data possibly being outdated by up to five years or more. As these vehicles traverse roads rather than pedestrian pathways, there's a chance certain signages are overlooked, leading to incomplete or outdated data and potentially inaccurate results.

Concerning data analysis, while ArcGIS Pro generates the shortest path, the accuracy of minor results may not align with real-world scenarios. In practice, individuals may choose alternative routes different from those suggested by ArcGIS Pro. Moreover, the results of the average nearest neighbour method are limited as it summarizes the entire pattern using a single value. Furthermore, this method's outcome is heavily influenced by the study area's size, with larger areas potentially indicating clustering while smaller ones may reveal dispersion (ESRI, 2013).

Suggestions

Upon completion of this project, a self-review was conducted to provide suggestions for future research, whether academic or policy oriented. When conducting similar research in the future to assess the spatial analysis of desired study station and its directional signage placement, it is recommended to examine each studied city and station individually. The quest for providing improved solutions for signage placement and distribution cannot yield identical results across different contexts. Hence, it is imperative to approach each case separately. This means that a solution effective for one station may not be directly applicable to others, even within the same city. Attempting to generalize solutions across contexts may lead to misallocation of resources and compromise the quality of results.

Comparing between cities and stations is recommended in order to conduct a more comprehensive analysis. If only focusing on frequently visited tourist stations, aspects related to less frequently visited residential area stations may be overlooked for analysis and consideration. This oversight might hinder the improvement of the navigation system for various types of communities. It is important to note that while residential area stations may not require as much attention as representative stations, those less frequented should not be neglected.

In our study, the quantity of signs and their associated shortest travel distance and time, as well as coverage area, are primarily based on the buffer zone set at 500 meters. However, the choice of a 500-meter radius is subject to debate and may not provide a definitive answer for future applications, although a buffer zone is deemed necessary. It is suggested that the radius range from 100 meters up to 700 meters, depending on the background of the studied station.

A significant spatial parameter that could be considered for investigation is the observable or visible area of each directional signage by human sight. Essentially, it examines how far a person can see a signage from different angles, and assesses whether there is sufficient lighting at each signage position to determine if each signage is easily visible. However, this presents a formidable challenge. Firstly, it is a subjective determination, as the visible area varies among individuals. Secondly, the visibility of a signage can differs depending on the direction, with the north side potentially visible for 5 meters while the south side might be obstructed by buildings, rendering the observable area inconsistent from every perspective. Consequently, a simple circular buffer zone cannot adequately represent this variability. Moreover, collecting this data is a time-consuming process. It requires physical examination of every signage in cities like Hong Kong, Auckland, and London, which are globally distributed, by real individuals on-site, adding to the complexity and challenge of the task.

Another key and fundamental parameter that is highly recommended for spatial analysis is the orientation or true directional alignment of directional signages (Dogu & Erkip, 2000). This parameter was not included in the current study. Understanding this aspect can help determine whether the signage provides accurate and efficient directions for the shortest path.

Additionally, considering the geographically weight of signage as a parameter for analysis is crucial (Brunsdon et al., 1996). Certain streets may require more signage due to specific needs, while others may not have signage installed yet due to various factors, thus affecting the geographically weight of the signage. Geographically weight could be allocated based on major intersections, and comparing the weight scale to other locations would facilitate accessing the spatial results effectively.

Additionally, it's highly recommended to consider spatial autocorrelation (Grekousis, 2020) as a method for future research. This method measures the degree of spatial dependency, which can provide valuable insights for future study. Spatial autocorrelation can serve as the conceptual framework for exploring the spatial impact of directional train signages in each city, as well as identifying potential relationships. Furthermore, comparing the underlying spatial factors that affect spatial autocorrelation results between cities can help investigate any spatial impact on station signage. Instead of relying solely on statistical tests like Global Moran's I, conducting a simple analysis of the degree of spatial dependency and its underlying reasons is recommended. This approach allows for a more nuanced understanding of spatial relationships and their implications. To delve into why certain cities exhibit differences in spatial factors and how these factors influence the arrangement, quantity, or coverage area of signages, addressing the following questions is suggested.

- Do more developed cities build a greater number of signages or station entrances?
- Does the size of cities and stations influence the coverage area of signage?
- Does the density of cities impact signage distribution?
- Do the population sizes of cities influence signage placement?
- Do local residents' transportation habits affect the quantity of signage?
- Does the frequency and number of visitors impact signage quantity, considering tourism and visitor attractions?
- Does the older history of the train station company influence the quantity and spatial distribution of signage?
- Do cities with more commercial and business districts exhibit differences in signage patterns?
- Do environmental considerations impact the quantity of signage?
- Do cultural events and festivals in each city influence signage, potentially serving to promote and guide people to these events, thus affecting the quantity and spatial pattern of signage?

Lastly, this project primarily focuses on the spatial aspect, which, while important, isn't fully comprehensive. Signages aren't solely designed for aesthetic spatial distribution on maps, they're intended for pedestrians navigating the streets. Therefore, it's crucial to consider both the psychological aspect that can help the community and pedestrians easily see signs from a distance (Sun et al., 2022; Espera et al., 2015; Meis & Kashima, 2017), and how useful they find them based on survey ratings. Also, the physical aspect, including sign height, colour, and the design of messages and logos (Calori & Eynden, 2015), needs to be heavily taken into consideration for an effective signage system.

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